FINITE MÖBIUS INTERPOLATION AND PRIME-POWER SPECTRAL IDENTITIES FOR EXPONENTIAL WALKS OVER THE NONTRIVIAL ZEROS OF THE RIEMANN ZETA FUNCTION

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Remark 0.1. This manuscript develops the analytic identities in full theorem–proof form. Standard and literature-based material is cited at the point of use. Original finite identities are proved directly. Heuristics and conjectures are explicitly labelled. Numerical evidence is confined to the final part, immediately before the conclusions, and is never used as a substitute for an analytic proof.

1. Introduction and separation of the two constructions

Two finite constructions are studied.

The first construction begins with the Möbius Dirichlet polynomial and a distinguished set of primes in the terminal half interval. Its central object is a meromorphic interpolation $\Phi_N(s)$ whose

value at $s = 0$ is -2, whose large positive real limit is $M(N)$, and whose exponential rate recovers the first squarefree integer after N.

The second construction begins with the phase walk

$$S_q(T) = \sum_{0<\gamma\le T} m(\gamma) e^{i\gamma \log q}, \qquad q > 1, \tag{1.1}$$

where the nontrivial zeros are written as

$$\rho = \beta + i\gamma \tag{1.2}$$

and $m(\gamma)$ is the total multiplicity at ordinate γ. The computation subtracts a smooth counting contribution and finitely many prime-power oscillators. Exact identities are derived for the resulting carrier coefficient, for its imaginary part, and for its comparison with a convergent signed prime-power invariant.

The two constructions share the reciprocal zeta function and prime-power support, but they answer different questions. Part I is finite and algebraic. Part II concerns exponential sums over zeta ordinates and the explicit-formula interface.

Part 1. Finite Möbius interpolation

2. Terminal half-interval decomposition

Definition 2.1. For an integer $N \ge 2$, define

$$Q_N = \left\{ q \text{ prime} : \frac{N}{2} < q \le N \right\}, \tag{2.1}$$

and put

$$h_N = |Q_N|, \qquad q_N = \min Q_N. \tag{2.2}$$

Bertrand's postulate guarantees $h_N \ge 1$.

Definition 2.2. Let

$$M_s(N) = \sum_{n\le N} \frac{\mu(n)}{n^s}, \tag{2.3}$$

$$P_N(s) = \sum_{q\in Q_N} q^{-s}, \tag{2.4}$$

$$A_N(s) = \sum_{\substack{n\le N \\ n\notin Q_N}} \frac{\mu(n)}{n^s}, \tag{2.5}$$

$$B_N = A_N(0) = \sum_{\substack{n\le N \\ n\notin Q_N}} \mu(n). \tag{2.6}$$

Since every $q \in Q_N$ is prime and $\mu(q) = -1$,

$$M_s(N) = A_N(s) - P_N(s) \tag{2.7}$$

and

$$M(N) = B_N - h_N. \tag{2.8}$$

Definition 2.3. For $\Re s > 1$, define the exact tail

$$R_N(s) = \frac{1}{\zeta(s)} - M_s(N) = \sum_{n>N} \frac{\mu(n)}{n^s}. \tag{2.9}$$

The first equality supplies a meromorphic continuation of R_N.

Definition 2.4. Define

$$\Phi_N(s) = B_N + h_N \frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)} \tag{2.10}$$

at points at which the quotient is defined, and meromorphically elsewhere.

3. Canonical identity and endpoint values

Theorem 3.1 (Canonical orientation)**.** *For every $N \geq 2$,*

$$\boxed{\Phi_N(s) = B_N + h_N \frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)} = M(N) + h_N \frac{R_N(s)}{P_N(s)}.} \tag{3.1}$$

Proof. From (2.7) and (2.9),

$$\zeta(s)^{-1} - A_N(s) = R_N(s) + M_s(N) - A_N(s) \tag{3.2}$$

$$= R_N(s) - P_N(s). \tag{3.3}$$

Substitution into (2.10), followed by (2.8), gives

$$\Phi_N(s) = B_N + h_N \frac{R_N(s) - P_N(s)}{P_N(s)} \tag{3.4}$$

$$= B_N - h_N + h_N \frac{R_N(s)}{P_N(s)} \tag{3.5}$$

$$= M(N) + h_N \frac{R_N(s)}{P_N(s)}. \tag{3.6}$$

□

Corollary 3.2 (Value at the origin)**.** *The meromorphic interpolation satisfies*

$$\boxed{\Phi_N(0) = -2.} \tag{3.7}$$

Proof. The classical value $\zeta(0) = -1/2$ gives $\zeta(0)^{-1} = -2$. Moreover, $A_N(0) = B_N$ and $P_N(0) = h_N$. Therefore

$$\Phi_N(0) = B_N + h_N \frac{-2 - B_N}{h_N} \tag{3.8}$$

$$= -2. \tag{3.9}$$

□

Corollary 3.3 (Large positive real limit)**.** *Along the positive real axis,*

$$\boxed{\lim_{\sigma \to +\infty} \Phi_N(\sigma) = M(N).} \tag{3.10}$$

Proof. By (2.9), $R_N(\sigma)$ is supported on integers $n > N$. The denominator $P_N(\sigma)$ is supported on integers $q \leq N$. Consequently $R_N(\sigma)/P_N(\sigma) \to 0$, and (3.6) gives the result. □

Experiment 3.4 (The $N = 10$ orientation check)**.** For $N = 10$,

$$Q_{10} = \{7\}, \qquad h_{10} = 1, \qquad M(10) = -1, \qquad B_{10} = 0. \tag{3.11}$$

The inverted quotient P_N/R_N would give $-1/2$ at $s = 0$. The quotient in (3.1) gives

$$0 + \frac{-2}{1} = -2. \tag{3.12}$$

This finite check confirms the orientation of the quotient in Theorem 3.1.

4. Exponential approach, rounding, and recovery of the first tail term

Definition 4.1. Let

$$a_N = \min\{n > N : \mu(n) \neq 0\}. \tag{4.1}$$

Theorem 4.2 (Leading exponential term)**.** *As $\sigma \to +\infty$,*

$$\boxed{\Phi_N(\sigma) - M(N) = h_N \mu(a_N) \left(\frac{q_N}{a_N}\right)^{\sigma} (1 + o(1)).} \tag{4.2}$$

Proof. The least element of Q_N is q_N, so

$$P_N(\sigma) = q_N^{-\sigma} (1 + o(1)). \tag{4.3}$$

By the definition of a_N,

$$R_N(\sigma) = \mu(a_N) a_N^{-\sigma} (1 + o(1)). \tag{4.4}$$

Insert (4.3) and (4.4) into (3.6). □

Remark 4.3. The factor is $(q_N/a_N)^{\sigma}$, not $(a_N/q_N)^{\sigma}$. Since $q_N \leq N < a_N$, the correct factor decays. The inverted factor would contradict Corollary 3.3.

Theorem 4.4 (Uniform real-axis error bound)**.** *For every real $\sigma > 1$,*

$$\boxed{|\Phi_N(\sigma) - M(N)| \leq \frac{h_N N}{\sigma - 1} \left(\frac{q_N}{N}\right)^{\sigma}.} \tag{4.5}$$

Proof. Absolute convergence for $\sigma > 1$ gives

$$|R_N(\sigma)| \leq \sum_{n>N} n^{-\sigma} \tag{4.6}$$

$$\leq \int_N^{\infty} x^{-\sigma} \, \mathrm{d}x = \frac{N^{1-\sigma}}{\sigma - 1}. \tag{4.7}$$

Since $q_N \in Q_N$,

$$P_N(\sigma) \geq q_N^{-\sigma}. \tag{4.8}$$

The result follows from (3.6). □

Corollary 4.5 (Finite rounding criterion)**.** *If*

$$\boxed{\frac{h_N N}{\sigma - 1} \left(\frac{q_N}{N}\right)^{\sigma} < \frac{1}{2},} \tag{4.9}$$

then $M(N)$ is the unique nearest integer to the real number $\Phi_N(\sigma)$.

Definition 4.6. Define

$$E_N(s) = \frac{\Phi_N(s) - M(N)}{h_N} = \frac{R_N(s)}{P_N(s)}. \tag{4.10}$$

Corollary 4.7 (Recovery of a_N)**.** *The first nonzero Möbius term after N is recovered by*

$$\boxed{a_N = q_N \lim_{\sigma \to +\infty} |E_N(\sigma)|^{-1/\sigma}.} \tag{4.11}$$

Proof. Theorem 4.2 gives

$$|E_N(\sigma)|^{-1/\sigma} \longrightarrow \frac{a_N}{q_N}. \tag{4.12}$$

□

5. Feasible terminal values and non-distinction in $\Re s > 1$

Definition 5.1. For an integer parameter m, define

$$F_N(s;m) = M_s(N) + P_N(s)\left(1 + \frac{m - B_N}{h_N}\right). \tag{5.1}$$

A value m is called terminally feasible when

$$-1 \le \frac{m - B_N}{h_N} \le 1. \tag{5.2}$$

Proposition 5.2 (Exact substitution identity)**.** *The expression* (5.1) *is the exact result of substituting*

$$A_N(s) = M_s(N) + P_N(s) \tag{5.3}$$

into the terminal-value interpolation.

Theorem 5.3 (The half-plane limit does not select $M(N)$)**.** *Fix s with $\Re s > 1$. For every sequence of feasible integers $m(N)$,*

$$\boxed{F_N(s;m(N)) \longrightarrow \frac{1}{\zeta(s)} \qquad (N \to \infty).} \tag{5.4}$$

Thus convergence in the absolutely convergent half-plane does not distinguish $m(N) = M(N)$ from other feasible terminal values.

Proof. For $\Re s > 1$,

$$M_s(N) \longrightarrow \frac{1}{\zeta(s)} \tag{5.5}$$

by absolute convergence. Since $Q_N \subset (N/2, N]$,

$$|P_N(s)| \le \sum_{N/2 < n \le N} n^{-\Re s} \longrightarrow 0. \tag{5.6}$$

Feasibility implies that the factor in parentheses in (5.1) lies in $[0, 2]$. Its product with $P_N(s)$ therefore tends to zero. □

Remark 5.4. The selection of the true value $M(N)$ is arithmetical and combinatorial. It is not produced by convergence to $1/\zeta(s)$ in $\Re s > 1$, because the entire feasible family has the same limit.

6. Poles and finite parts at zeta zeros

Definition 6.1. If a meromorphic function has a Laurent expansion

$$f(s) = \frac{a_{-1}}{s - \rho} + a_0 + O(s - \rho), \tag{6.1}$$

its finite part at ρ is

$$\mathrm{FP}_{s=\rho}\, f(s) = a_0. \tag{6.2}$$

Proposition 6.2 (Finite part of the reciprocal zeta function)**.** *Let ρ be a simple zero of ζ. Then*

$$\frac{1}{\zeta(s)} = \frac{1}{\zeta'(\rho)(s - \rho)} - \frac{\zeta''(\rho)}{2\zeta'(\rho)^2} + O(s - \rho), \tag{6.3}$$

and hence

$$\mathrm{FP}_{s=\rho}\, \frac{1}{\zeta(s)} = -\frac{\zeta''(\rho)}{2\zeta'(\rho)^2}. \tag{6.4}$$

Proof. Expand $\zeta(s)$ at its simple zero and invert the resulting power series. □

Proposition 6.3 (Finite part of Φ_N)**.** *Assume that ρ is a simple zero of ζ and $P_N(\rho) \neq 0$. Then*

$$\mathrm{FP}_{s=\rho}\,\Phi_N(s) = B_N + h_N \frac{-\dfrac{\zeta''(\rho)}{2\zeta'(\rho)^2} - A_N(\rho)}{P_N(\rho)} \tag{6.5}$$

$$- h_N \frac{P_N'(\rho)}{\zeta'(\rho) P_N(\rho)^2}. \tag{6.6}$$

Proof. Write $z = s - \rho$. Proposition 6.2 gives

$$\zeta(s)^{-1} - A_N(s) = \frac{1}{\zeta'(\rho) z} + b_0 + O(z), \tag{6.7}$$

where

$$b_0 = -\frac{\zeta''(\rho)}{2\zeta'(\rho)^2} - A_N(\rho). \tag{6.8}$$

Also

$$P_N(s) = P_N(\rho) + P_N'(\rho) z + O(z^2). \tag{6.9}$$

Division of the two expansions yields the stated constant term. □

Definition 6.4. For a simple zero ρ, define the normalized finite-part observable

$$\mathcal{F}_N(\rho) = N^{-\rho}\,\mathrm{FP}_{s=\rho}\,\Phi_N(s). \tag{6.10}$$

Experiment 6.5 (Scale scan)**.** A finite scan over

$$N = 2000, 4000, 8000, 16000, 32000, 64000 \tag{6.11}$$

produced successive magnitudes

$$|\mathcal{F}_{2N}(\rho) - \mathcal{F}_N(\rho)| \approx 1.12,\ 0.75,\ 0.80,\ 0.75,\ 0.44. \tag{6.12}$$

The differences do not decrease monotonically and remain of the same order as the values themselves. With six scales this is a negative finite result: no scale cancellation law is established.

Experiment 6.6 (Algebraic finite-part audit)**.** In the numerical implementation, the direct tail expression and the finite-part reconstruction agreed to

$$|R_{\text{from FP}} - R_{\text{direct}}| = 4.043 \times 10^{-10}. \tag{6.13}$$

This checks the algebra and numerical implementation; it is not an asymptotic result.

Part 2. Prime-power phase walks over zeta ordinates

7. Phase walks, endpoint conventions, and a universal half

Definition 7.1. Fix $q > 1$ and write

$$\omega = \log q. \tag{7.1}$$

For the positive ordinates, counted with multiplicity, define

$$S_q(T) = \sum_{0<\gamma\le T} m(\gamma) e^{i\omega\gamma}. \tag{7.2}$$

At an ordinate γ_k, the right, left, and midpoint values are

$$S_k^{\text{right}} = \sum_{j\le k} b_j, \tag{7.3}$$

$$S_k^{\text{left}} = \sum_{j<k} b_j, \tag{7.4}$$

$$S_k^{\mathrm{mid}} = \sum_{j<k} b_j + \frac{1}{2} b_k, \qquad b_k = e^{i\omega\gamma_k}. \tag{7.5}$$

Proposition 7.2 (Endpoint conversion)**.** *At every index k,*

$$\overline{b_k} S_k^{\mathrm{right}} = \overline{b_k} S_k^{\mathrm{mid}} + \frac{1}{2}, \tag{7.6}$$

$$\overline{b_k} S_k^{\mathrm{left}} = \overline{b_k} S_k^{\mathrm{mid}} - \frac{1}{2}. \tag{7.7}$$

Thus changing the endpoint convention translates the carrier coefficient by $\pm 1/2$ and does not change a centered variance.

Theorem 7.3 (Exact triangular half identity)**.** *Let $|b_k| = 1$ and*

$$S_k = \sum_{j \le k} b_j. \tag{7.8}$$

Then

$$\boxed{\Re\left(\frac{1}{N}\sum_{k=1}^{N} \overline{b_k} S_k\right) = \frac{1}{2} + \frac{|S_N|^2}{2N}.} \tag{7.9}$$

Proof. Let L be the lower triangular $N \times N$ matrix with entries $L_{kj} = 1$ for $j \le k$, and let J be the all-ones matrix. Then

$$L + L^* = J + I. \tag{7.10}$$

For $b = (b_1, \dots, b_N)^{\mathsf{T}}$,

$$2\Re(b^* L b) = b^*(J + I)b \tag{7.11}$$

$$= \left|\sum_{k=1}^{N} b_k\right|^2 + N. \tag{7.12}$$

Divide by $2N$. □

Corollary 7.4. *For midpoint evaluation,*

$$\Re\left(\frac{1}{N}\sum_{k=1}^{N} \overline{b_k} S_k^{\mathrm{mid}}\right) = \frac{|S_N|^2}{2N}. \tag{7.13}$$

Warning 7.5. Theorem 7.3 is universal for unit phases. It does not identify prime-power structure and does not imply the Riemann Hypothesis. It applies exactly to the raw full-sample cumulative walk, not automatically to a residual after smooth and prime-power subtraction.

8. Complete-zero weights and the positive horizontal defect

Definition 8.1. For a positive ordinate τ, define

$$W_q(\tau) = \sum_{\rho : \Im\rho = \tau} q^{\beta - \frac{1}{2}}, \tag{8.1}$$

where zeros are counted with multiplicity.

Definition 8.2. Define

$$D_q(\tau) = \sum_{\substack{\rho = \beta + i\tau \\ \beta > 1/2}} \left[\cosh\left((\beta - \tfrac{1}{2})\log q\right) - 1\right]. \tag{8.2}$$

Theorem 8.3 (Positive defect at each ordinate)**.** *For every $q > 1$ and every positive ordinate τ,*

$$\boxed{W_q(\tau) = m(\tau) + 2D_q(\tau), \qquad D_q(\tau) \geq 0.} \tag{8.3}$$

Moreover,

$$D_q(\tau) = 0 \tag{8.4}$$

if and only if every zero at ordinate τ lies on the critical line.

Proof. The functional equation and conjugation symmetry pair every [11, 5] $\beta + i\tau$ with $1 - \beta + i\tau$. A pair with $\delta = \beta - 1/2 > 0$ contributes

$$q^{\delta} + q^{-\delta} = 2\cosh(\delta \log q) \tag{8.5}$$

$$= 2 + 2\left[\cosh(\delta \log q) - 1\right]. \tag{8.6}$$

Zeros on the critical line contribute one each. Since $\cosh x - 1 \geq 0$, with equality only at $x = 0$, the result follows. □

Definition 8.4. Define the locally finite complex measures

$$\mu_q = \sum_{\tau > 0} m(\tau) e^{i\tau \log q} \delta_\tau, \tag{8.7}$$

$$\nu_q = \sum_{\tau > 0} W_q(\tau) e^{i\tau \log q} \delta_\tau. \tag{8.8}$$

Corollary 8.5 (Atomic positive criterion)**.** *The exact measure identity*

$$\boxed{\frac{1}{2} e^{-iT \log q}(\nu_q - \mu_q) = \sum_{\tau > 0} D_q(\tau) \delta_\tau} \tag{8.9}$$

holds. For any one fixed $q > 1$,

$$\nu_q = \mu_q \quad \Longleftrightarrow \quad \text{RH}. \tag{8.10}$$

Proof. Equation (8.9) is Theorem 8.3 applied atom by atom. A positive atomic measure is zero if and only if every atomic mass is zero. □

Remark 8.6. No value of $q > 1$ is logically privileged. For a small displacement $\delta = \beta - 1/2$,

$$\cosh(\delta \log q) - 1 = \frac{1}{2}\delta^2 (\log q)^2 + O\left(\delta^4 (\log q)^4\right). \tag{8.11}$$

Larger q amplifies small horizontal displacements but also enlarges the analytic weights.

9. The explicit-formula origin of the two prime-power branches

The Guinand explicit formula and its later formulations relate suitable test-function sums over zeta zeros to the von Mangoldt function $\Lambda(n)$, hence to prime powers [9, 12]. Landau's formula and Gonek's uniform refinements isolate the same prime-power resonances in exponential sums over zeros [8, 7, 6].

Definition 9.1. Let $n = p^r$ be a prime power, put

$$\ell_n = \log n, \tag{9.1}$$

and suppose $\ell_n \neq \omega$. Define

$$c_n^- = \frac{i\Lambda(n)}{2\pi\sqrt{n}(\omega - \ell_n)}, \tag{9.2}$$

$$c_n^+ = \frac{i\Lambda(n)}{2\pi\sqrt{n}(\omega + \ell_n)}. \tag{9.3}$$

The associated oscillators are

$$P_n^-(T) = c_n^- e^{i(\omega-\ell_n)T}, \tag{9.4}$$

$$P_n^+(T) = c_n^+ e^{i(\omega+\ell_n)T}. \tag{9.5}$$

Heuristic 9.2 (Sharp-window prime-spectral ansatz)**.** For a sharp counting window, formal integration of the explicit formula suggests a decomposition

$$S_q(T) = C_q + A_{\mathrm{smooth},q}(T) + \frac{1}{2}e^{i\omega T} + \sum_{n=p^r}\left(P_n^-(T) + P_n^+(T)\right), \tag{9.6}$$

where the infinite prime-power sum requires a specified truncation or regularization. This is a heuristic argument when used with the discontinuous indicator $\mathbf{1}_{\gamma\le T}$.

Definition 9.3 (Smooth term used in the computation)**.** The MATLAB program uses

$$A_{\mathrm{smooth}}(T) = \frac{e^{i\omega T}\log(T/2\pi) - \mathrm{Ei}(i\omega T)}{2\pi i\omega}, \tag{9.7}$$

up to a branch-dependent additive constant. The constant is treated by a joint constant–carrier fit.

Warning 9.4. Equation (9.7) is the antiderivative of the smooth Riemann–von Mangoldt density multiplied by $e^{i\omega T}$. It is not, by itself, a complete sharp-window explicit formula. Gamma-factor, pole, endpoint, branch, and regularization conventions must be retained when an exact global identity is claimed.

10. Exact algebra of each prime-power mode

Proposition 10.1 (Sum, difference, and ratio identities)**.** *For every nonresonant prime power n,*

$$c_n^- + c_n^+ = \frac{i\Lambda(n)\omega}{\pi\sqrt{n}(\omega^2-\ell_n^2)}, \tag{10.1}$$

$$c_n^- - c_n^+ = \frac{i\Lambda(n)\ell_n}{\pi\sqrt{n}(\omega^2-\ell_n^2)}, \tag{10.2}$$

$$\frac{c_n^-}{c_n^+} = \frac{\omega+\ell_n}{\omega-\ell_n}. \tag{10.3}$$

Proof. Each identity follows by combining the two rational denominators in (9.2) and (9.3). □

Definition 10.2. Write

$$c_n^- = ia_n^-, \qquad c_n^+ = ia_n^+, \qquad a_n^\pm \in \mathbb{R}, \tag{10.4}$$

and define

$$u_n = a_n^- + a_n^+, \tag{10.5}$$

$$v_n = a_n^- - a_n^+, \tag{10.6}$$

$$d_n = |c_n^-|^2 - |c_n^+|^2. \tag{10.7}$$

Proposition 10.3 (Signed mode invariant)**.** *The quantities in Definition 10.2 satisfy*

$$u_n = -\frac{\omega\Lambda(n)}{\pi\sqrt{n}(\ell_n^2-\omega^2)}, \tag{10.8}$$

$$v_n = -\frac{\Lambda(n)\ell_n}{\pi\sqrt{n}(\ell_n^2-\omega^2)}, \tag{10.9}$$

$$\boxed{d_n = u_n v_n = \frac{\omega\Lambda(n)^2\ell_n}{\pi^2 n(\ell_n^2-\omega^2)^2}.} \tag{10.10}$$

Proof. The formulas for u_n and v_n are the imaginary coefficients in (10.1) and (10.2). Since $c_n^{\pm} = ia_n^{\pm}$,

$$|c_n^-|^2 - |c_n^+|^2 = (a_n^-)^2 - (a_n^+)^2 = u_n v_n. \tag{10.11}$$

□

Proposition 10.4 (Elliptic form and anisotropy)**.** *After demodulation by $e^{-i\omega T}$, the mode is*

$$g_n(T) = c_n^- e^{-i\ell_n T} + c_n^+ e^{i\ell_n T} = \frac{\Lambda(n)}{\pi\sqrt{n}(\omega^2 - \ell_n^2)} \left[\ell_n \sin(\ell_n T) + i\omega \cos(\ell_n T)\right]. \tag{10.12}$$

Its unsigned elliptic area is

$$\pi d_n. \tag{10.13}$$

For uniform phase, its anisotropy is

$$\mathcal{A}_n = \frac{\ell_n^2 - \omega^2}{\ell_n^2 + \omega^2} \longrightarrow 1 \qquad (n \to \infty). \tag{10.14}$$

Proof. Expanding the exponentials gives (10.12). The semiaxes have absolute product d_n, giving (10.13). The uniform-phase variances of the real and imaginary coordinates are proportional to ℓ_n^2 and ω^2, respectively, which gives (10.14). □

Lemma 10.5 (Dominance of $n = 3$ in the minus branch for $\omega = 1$)**.** *For $\omega = 1$, the largest coefficient $|c_n^-|$ among prime powers $n \geq 2$ occurs at $n = 3$.*

Proof. For $n \geq 3$,

$$|c_n^-| \leq \frac{\log n}{2\pi\sqrt{n}(\log n - 1)}. \tag{10.15}$$

The right-hand side is decreasing for $n > e$, and equality holds at a prime. Direct comparison with $n = 2$ completes the proof. □

11. The convergent signed tail

Definition 11.1. Define

$$D_\infty = \sum_{n=p^r} d_n, \tag{11.1}$$

$$\Delta_\omega(Q) = \sum_{\substack{n=p^r \\ n>Q}} d_n = D_\infty - \sum_{\substack{n=p^r \\ n\leq Q}} d_n. \tag{11.2}$$

Theorem 11.2 (Asymptotic signed tail)**.** *For fixed $\omega > 0$,*

$$\boxed{\Delta_\omega(Q) \sim \frac{\omega}{\pi^2 \log Q} \qquad (Q \to \infty).} \tag{11.3}$$

Proof. For primes p,

$$d_p = \frac{\omega}{\pi^2 p \log p} \left(1 - \frac{\omega^2}{(\log p)^2}\right)^{-2} \tag{11.4}$$

$$= \frac{\omega}{\pi^2 p \log p} + O_\omega\left(\frac{1}{p(\log p)^3}\right). \tag{11.5}$$

The prime number theorem and partial summation give [1, 2]

$$\sum_{p>Q} \frac{1}{p \log p} \sim \int_Q^\infty \frac{\mathrm{d}x}{x(\log x)^2} = \frac{1}{\log Q}. \tag{11.6}$$

For proper powers p^r, $r \geq 2$, the corresponding sum is $O_\omega(Q^{-1/2})$. Combining the two contributions proves the claim. □

Remark 11.3. For $\omega = 1$,

$$\pi\Delta_1(Q) \sim \frac{1}{\pi \log Q}, \tag{11.7}$$

$$\frac{3}{4}\Delta_1(Q) \sim \frac{3}{4\pi^2 \log Q}. \tag{11.8}$$

These constants match two finite numerical quantities, but the matching relations themselves remain conjectural.

12. Finite zero-sampled carrier identities

Definition 12.1. Fix a finite multiset of positive ordinates

$$\Gamma_H = \{\gamma_k : 0 < \gamma_k \leq H\}, \qquad N(H) = |\Gamma_H|, \tag{12.1}$$

and let Y be any complex-valued function on Γ_H. For a cutoff Q, define

$$R_Q(\gamma) = Y(\gamma) - \sum_{\substack{n=p^r \\ n\leq Q}} \left(P_n^-(\gamma) + P_n^+(\gamma)\right). \tag{12.2}$$

The raw carrier coefficient and zero kernel are

$$\widehat{c}_{Q,H} = \frac{1}{N(H)} \sum_{\gamma\in\Gamma_H} e^{-i\omega\gamma} R_Q(\gamma), \tag{12.3}$$

$$M_H(x) = \frac{1}{N(H)} \sum_{\gamma\in\Gamma_H} x^{i\gamma}, \qquad x > 0. \tag{12.4}$$

Also put

$$\widehat{c}_{Y,H} = \frac{1}{N(H)} \sum_{\gamma\in\Gamma_H} e^{-i\omega\gamma} Y(\gamma). \tag{12.5}$$

Theorem 12.2 (Exact finite-sample carrier formula)**.** *For the objects in Definition 12.1,*

$$\boxed{\widehat{c}_{Q,H} = \widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\leq Q}} \left(c_n^- \overline{M_H(n)} + c_n^+ M_H(n)\right).} \tag{12.6}$$

Proof. The demodulated branch identities are

$$e^{-i\omega\gamma} P_n^-(\gamma) = c_n^- n^{-i\gamma}, \tag{12.7}$$

$$e^{-i\omega\gamma} P_n^+(\gamma) = c_n^+ n^{i\gamma}. \tag{12.8}$$

Average (12.2) over Γ_H. □

Theorem 12.3 (Exact imaginary-part formula)**.** *Write*

$$M_H(n) = x_n + iy_n. \tag{12.9}$$

Then

$$\Im\left(c_n^- \overline{M_H(n)} + c_n^+ M_H(n)\right) = u_n x_n, \tag{12.10}$$

and consequently

$$\boxed{\Im\widehat{c}_{Q,H} = \Im\widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\leq Q}} u_n \Re M_H(n).} \tag{12.11}$$

Proof. Since $c_n^{\pm} = ia_n^{\pm}$,

$$c_n^- \overline{M_H(n)} + c_n^+ M_H(n) = ia_n^-(x_n - iy_n) + ia_n^+(x_n + iy_n) \tag{12.12}$$

$$= v_n y_n + iu_n x_n. \tag{12.13}$$

Insert this identity into Theorem 12.2. □

Definition 12.4. Define the Q-independent quantity

$$C_H = \Im \widehat{c}_{Y,H} - \pi D_\infty. \tag{12.14}$$

Theorem 12.5 (Exact global defect identity)**.** *For the finite-sample raw coefficient,*

$$\boxed{\Im \widehat{c}_{Q,H} - \pi \Delta_\omega(Q) = C_H - \sum_{\substack{n=p^r \\ n \le Q}} u_n \left[\Re M_H(n) - \pi v_n\right].} \tag{12.15}$$

Proof. By Theorem 12.3,

$$\Im \widehat{c}_{Q,H} = \Im \widehat{c}_{Y,H} - \sum_{n \le Q} u_n \Re M_H(n). \tag{12.16}$$

By Definitions 11.1 and 10.2,

$$\pi \Delta_\omega(Q) = \pi D_\infty - \pi \sum_{n \le Q} u_n v_n. \tag{12.17}$$

Subtract and use (12.14). □

Corollary 12.6 (Equivalent transport target)**.** *The conjectural relation*

$$\Im \widehat{c}_{Q,H} = \pi \Delta_\omega(Q) + o\left(\frac{1}{\log Q}\right) \tag{12.18}$$

is exactly equivalent to

$$\boxed{C_H - \sum_{n \le Q} u_n \left[\Re M_H(n) - \pi v_n\right] = o\left(\frac{1}{\log Q}\right).} \tag{12.19}$$

Theorem 12.7 (Exact increment identity)**.** *When Q crosses a prime power n,*

$$\Delta \Im \widehat{c}_{Q,H} = -u_n \Re M_H(n), \tag{12.20}$$

$$\Delta\left(\pi \Delta_\omega(Q)\right) = -\pi u_n v_n. \tag{12.21}$$

If $u_n v_n \neq 0$, then

$$\boxed{\frac{\Delta \Im \widehat{c}_{Q,H}}{\Delta(\pi \Delta_\omega(Q))} = \frac{\Re M_H(n)}{\pi v_n}.} \tag{12.22}$$

Proof. Take the difference of the two cutoff sums immediately before and after the single prime power n. □

13. Landau scaling and failure of termwise area transport

Landau's formula concerns sums of x^ρ, not sums of $x^{i\gamma}$ without horizontal weights. Passing from the former to the latter requires RH, or a separate control of the real parts. Uniformity when x grows with H requires a uniform Landau–Gonek formula [8, 7].

Heuristic 13.1 (Conditional Landau estimate for the ordinate kernel)**.** Assume RH and a uniform Landau–Gonek estimate in the range under consideration. With

$$L_H = \log(H/2\pi), \tag{13.1}$$

one expects at a prime power n

$$\Re M_H(n) \approx -\frac{\Lambda(n)}{\sqrt{n}\, L_H}. \tag{13.2}$$

Proposition 13.2 (Required termwise target)**.** *Individual transport of the signed area of mode n would require*

$$\Re M_H(n) \sim \pi v_n. \tag{13.3}$$

For a prime p,

$$\pi v_p = -\frac{(\log p)^2}{\sqrt{p}((\log p)^2 - \omega^2)} \sim -\frac{1}{\sqrt{p}}. \tag{13.4}$$

Heuristic 13.3 (Termwise ratio)**.** Combining (13.2) and (13.4) gives

$$\frac{\Re M_H(p)}{\pi v_p} \approx \frac{\log p}{L_H}. \tag{13.5}$$

If $p \asymp H^\theta$, this predicts

$$\frac{\Re M_H(p)}{\pi v_p} \approx \theta. \tag{13.6}$$

Thus termwise transport cannot be asymptotically exact at a fixed $\theta < 1$; it becomes plausible only for $p = H^{1-o(1)}$.

Warning 13.4. The fixed-p Landau formula alone does not prove (13.6). The latter is a falsifiable scaling prediction that requires uniformity in the joint range $p = p(H)$.

Conjecture 13.5 (Local transport near the height scale)**.** *Let*

$$Q_1, Q_2 = H^{1-o(1)}. \tag{13.7}$$

In a suitable uniform regime,

$$\Im \widehat{c}_{Q_2,H} - \Im \widehat{c}_{Q_1,H} \sim \pi \left[\Delta_\omega(Q_2) - \Delta_\omega(Q_1)\right]. \tag{13.8}$$

Proposition 13.6 (Exact local defect)**.** *The defect in Conjecture 13.5 is exactly*

$$\begin{aligned} &[\Im \widehat{c}_{Q_2,H} - \Im \widehat{c}_{Q_1,H}] - \pi \left[\Delta_\omega(Q_2) - \Delta_\omega(Q_1)\right] \\ &\quad = -\sum_{Q_1 < n \le Q_2} u_n \left[\Re M_H(n) - \pi v_n\right]. \end{aligned} \tag{13.9}$$

Proof. Subtract (12.15) at Q_1 and Q_2. The constant C_H cancels. □

14. Mixed carrier area versus prime self-area

Proposition 14.1 (Mixed form measured by the centroid coefficient)**.** *The imaginary part of the raw carrier coefficient is*

$$\Im \widehat{c}_{Q,H} = \frac{1}{N(H)} \sum_{\gamma_k \le H} \Im \left[e^{-i\omega\gamma_k} R_Q(\gamma_k)\right] \tag{14.1}$$

$$= -\frac{1}{N(H)} \sum_{\gamma_k \le H} \Im \left[\overline{R_Q(\gamma_k)} e^{i\omega\gamma_k}\right]. \tag{14.2}$$

It is a mixed bilinear pairing between the residual position and the actual phase increment.

Definition 14.2. For a finite prime-power model G, define the formal self-area

$$\mathfrak{A}_H(G) = -\frac{\pi}{H}\int \Im\left(\overline{G(t)}\,\mathcal{J}G(t)\right)\,\mathrm{d}t, \tag{14.3}$$

where $\mathcal{J}$ is a specified frequency-sign or Hilbert-type operator.

Warning 14.3. The mixed pairing (14.2) is not identical to the self-pairing (14.3). Such an identification would require a justified replacement

$$e^{i\omega\gamma_k} \longleftrightarrow \mathcal{J}G_Q(\gamma_k). \tag{14.4}$$

The exact termwise defect of this replacement is encoded by $\Re M_H(n) - \pi v_n$.

15. Exact quadratic form for the sampled variance

Definition 15.1 (Sampled inner product and removed Gram matrix)**.** For functions on Γ_H, define

$$\langle f, g\rangle_H = \frac{1}{N(H)}\sum_{\gamma_k\le H}\overline{f(\gamma_k)}g(\gamma_k). \tag{15.1}$$

Put

$$\mu_H = M_H(e^{\omega}) \tag{15.2}$$

and

$$G_H = \begin{pmatrix} 1 & \mu_H \\ \overline{\mu_H} & 1 \end{pmatrix}. \tag{15.3}$$

Definition 15.2. For each prime power n and sign $\sigma \in \{-1, +1\}$, define the sampled frequency vector

$$v_{n,\sigma}(\gamma_k) = n^{i\sigma\gamma_k}. \tag{15.4}$$

Proposition 15.3 (Exact unprojected Gram kernel)**.** *The sampled Gram entries are*

$$\langle v_{n,\sigma}, v_{m,\tau}\rangle_H = M_H\left(m^{\tau}n^{-\sigma}\right). \tag{15.5}$$

Thus the four branch interactions between p^r and q^s involve

$$M_H(q^s/p^r), \quad M_H(q^sp^r), \quad M_H((q^sp^r)^{-1}), \quad M_H(p^r/q^s). \tag{15.6}$$

Definition 15.4. Let Π_H be the orthogonal projection onto the complement of

$$\mathrm{span}\{1, e^{i\omega\gamma}\}. \tag{15.7}$$

Define the projected kernel

$$\widetilde{K}_H((n,\sigma),(m,\tau)) = \langle \Pi_H v_{n,\sigma}, \Pi_H v_{m,\tau}\rangle_H. \tag{15.8}$$

Proposition 15.5 (Schur-complement formula)**.** *Let*

$$K_H(\lambda,\mu) = \frac{1}{N(H)}\sum_{\gamma_k\le H} e^{i(\mu-\lambda)\gamma_k}, \tag{15.9}$$

and

$$u_H(\lambda) = \begin{pmatrix} K_H(0,\lambda) \\ K_H(\omega,\lambda)\end{pmatrix}. \tag{15.10}$$

Then

$$\widetilde{K}_H(\lambda,\mu) = K_H(\lambda,\mu) - u_H(\lambda)^* G_H^{-1} u_H(\mu), \tag{15.11}$$

where G_H is given by (15.3)*.*

Proof. This is the standard Gram-matrix formula for orthogonal projection away from a finite-dimensional subspace. □

Theorem 15.6 (Exact system variance)**.** *Let a be the vector of all retained branch coefficients c_n^σ. The variance after joint removal of the constant and carrier is exactly*

$$\boxed{V(Q,H) = a^* \widetilde{K}_H a.} \tag{15.12}$$

Grouping by prime bases gives

$$V(Q,H) = \sum_p a_p^* \widetilde{K}_{p,p;H} a_p + \sum_{p \neq q} a_p^* \widetilde{K}_{p,q;H} a_q. \tag{15.13}$$

Warning 15.7 (The diagonal issue). The scalar Parseval diagonal

$$\sum_{p,r} \left(|c_{p^r}^-|^2 + |c_{p^r}^+|^2 \right) \tag{15.14}$$

is not the diagonal of the projected, zero-sampled system (15.12). Inside one prime tower, p^r/p^s and p^{r+s} are themselves prime powers, so the entire two-branch tower block is arithmetically resonant. It is currently unproved that tower blocks dominate, contract, or localize the residual.

Question 15.8. Determine the asymptotic size of

$$\mathfrak{E}_p(H) = a_p^* \widetilde{K}_{p,p;H} a_p \tag{15.15}$$

and the summed cross-tower contribution in (15.13), uniformly in a growing prime-power cutoff.

16. A correct shell calculation and its limited meaning

Definition 16.1. For fixed $a > 0$, define the finite multiplicative shell

$$\mathcal{S}_a(Q) = \{n = p^r : Q < n \leq e^a Q\} \tag{16.1}$$

and its demodulated sum

$$G_{Q,a}(T) = \sum_{n \in \mathcal{S}_a(Q)} g_n(T). \tag{16.2}$$

Proposition 16.2 (Bohr energy of a fixed logarithmic shell)**.** *For fixed $a > 0$,*

$$\lim_{X \to \infty} \frac{1}{X} \int_0^X |G_{Q,a}(T)|^2 \, \mathrm{d}T = \frac{a}{2\pi^2 \log Q} + o\left(\frac{1}{\log Q}\right) \tag{16.3}$$

as $Q \to \infty$.

Proof. Distinct frequencies $\pm \log n$ are orthogonal in Bohr mean. For primes,

$$|c_p^-|^2 + |c_p^+|^2 = \frac{1}{2\pi^2 p} + O_\omega\left(\frac{1}{p(\log p)^2}\right). \tag{16.4}$$

The prime number theorem and partial summation give [1, 2]

$$\sum_{Q < p \leq e^a Q} \frac{1}{p} = \frac{a}{\log Q} + o\left(\frac{1}{\log Q}\right). \tag{16.5}$$

Proper prime powers contribute a lower-order term. □

Corollary 16.3. *The choice $a = 3/2$ gives*

$$\frac{3}{4\pi^2 \log Q} + o\left(\frac{1}{\log Q}\right). \tag{16.6}$$

Warning 16.4. Corollary 16.3 proves only the Bohr energy of a deliberately selected hard shell. It does not prove that the actual residual is supported in $(Q, e^{3/2}Q]$. Many transfer profiles have the same quadratic mass, and the projected zero-sampled variance is not the Bohr norm in Proposition 16.2.

17. Regularization and the remaining explicit-formula bridge

Definition 17.1. A regularized counting window is a family $H_{T,\eta}(u)$ tending to the midpoint indicator of $[0,T]$ as $\eta \to \infty$. A formal Gaussian example is

$$H_{T,\eta}(u) = \int_0^T \frac{\eta}{\sqrt{\pi}} e^{-\eta^2(u-v)^2}\,\mathrm{d}v. \tag{17.1}$$

Heuristic 17.2 (Fourier displacement)**.** For

$$h_{T,\eta,\omega}(u) = e^{i\omega u} H_{T,\eta}(u), \tag{17.2}$$

formal Fourier transformation gives

$$\widehat{h}_{T,\eta,\omega}(v) = e^{-(v-\omega)^2/(4\eta^2)} \frac{1 - e^{-i(v-\omega)T}}{i(v-\omega)}. \tag{17.3}$$

Insertion into the explicit formula generates the two branches in Definition 9.1, multiplied by explicit regularizing weights.

Warning 17.3. To turn Section 17 into a theorem one must specify a test class for which the explicit formula is valid, retain every archimedean and boundary term, and control jointly:

$$H, \qquad Q, \qquad \eta. \tag{17.4}$$

A formal Gaussian manipulation is not a proof of a sharp-window formula.

Conjecture 17.4 (Uniform explicit formula for the ordinate walk)**.** *There exist choices $Q = Q(H) \to \infty$ and $\eta = \eta(H) \to \infty$ for which the regularized prime-power formula, including its complete archimedean and endpoint terms, converges locally to the midpoint ordinate walk, with an error tending to zero in a topology strong enough to identify the atomic measure* (8.9)*.*

Remark 17.5. The ordinary explicit formula reconstructs complete-zero weights, which retain $q^{\beta-1/2}$. Replacing that object by the unweighted ordinate walk is precisely where the positive defect $D_q(\tau)$ enters. Truncation control alone does not remove the horizontal zero information.

18. Energy exhaustion and distributional convergence

Conjecture 18.1 (Energy exhaustion)**.** *For a suitable joint regime $Q = Q(H) \to \infty$,*

$$\mathrm{Var}_{\gamma \le H}(F_{Q,H}) \le \varepsilon(Q,H), \qquad \varepsilon(Q(H),H) \to 0. \tag{18.1}$$

The variance is the projected discrete form (15.12)*, not an unprojected Lebesgue L^2-norm.*

Theorem 18.2 (Abstract local distributional implication)**.** *Suppose a sequence F_j satisfies*

$$F_j - \frac{1}{2} \longrightarrow 0 \quad \text{in } L^2_{\mathrm{loc}}(\mathbb{R}), \tag{18.2}$$

and

$$\frac{\mathrm{d}}{\mathrm{d}T}\left[e^{i\omega T}\left(F_j(T) - \frac{1}{2}\right)\right] \longrightarrow \nu_q - \mu_q \quad \text{in } \mathcal{D}'(\mathbb{R}). \tag{18.3}$$

Then RH *holds.*

Proof. Set

$$G_j(T) = e^{i\omega T}\left(F_j(T) - \frac{1}{2}\right). \tag{18.4}$$

Local L^2-convergence implies local L^1-convergence. Hence, for every compactly supported smooth φ,

$$\langle G_j', \varphi\rangle = -\int G_j(T)\varphi'(T)\,\mathrm{d}T \longrightarrow 0. \tag{18.5}$$

By (18.3), $\nu_q - \mu_q = 0$. Corollary 8.5 then gives RH. □

Warning 18.3. A normalized mean-square limit on intervals $[0, H]$ does not imply Theorem 18.2. A fixed compact defect, or even a sparse family of atomic defects, is diluted by division by H. Local or distributional convergence is indispensable.

19. Further analytic consequences of the finite Möbius interpolation

19.1. **Meromorphic structure and residues.**

Proposition 19.1 (Sources of poles)**.** *The meromorphic function Φ_N can have poles only at zeros of ζ and at zeros of the exponential polynomial P_N. At a simple zero ρ of ζ satisfying $P_N(\rho) \neq 0$, the residue is*

$$\operatorname*{Res}_{s=\rho} \Phi_N(s) = \frac{h_N}{\zeta'(\rho) P_N(\rho)}. \tag{19.1}$$

At a simple zero s_0 of P_N that is not a zero of ζ and satisfies $\zeta(s_0)^{-1} \neq A_N(s_0)$, the residue is

$$\operatorname*{Res}_{s=s_0} \Phi_N(s) = h_N \frac{\zeta(s_0)^{-1} - A_N(s_0)}{P_N'(s_0)}. \tag{19.2}$$

Proof. Both statements follow directly from the quotient in (2.10). At a simple zeta zero, $\zeta(s)^{-1}$ has principal part $1/(\zeta'(\rho)(s-\rho))$. At a simple zero of P_N, the denominator has principal part $P_N'(s_0)(s - s_0)$. □

Proposition 19.2 (Value at the pole of zeta)**.** *The reciprocal zeta function has a simple zero at $s = 1$, and therefore*

$$\Phi_N(1) = B_N - h_N \frac{A_N(1)}{P_N(1)}. \tag{19.3}$$

Proof. Use $\zeta(s)^{-1} \to 0$ as $s \to 1$ in (2.10). The denominator $P_N(1)$ is positive. □

19.2. **Eventually exact sign and quantitative rounding.**

Corollary 19.3 (Eventually exact sign)**.** *For sufficiently large real σ,*

$$\operatorname{sgn}\big(\Phi_N(\sigma) - M(N)\big) = \mu(a_N). \tag{19.4}$$

Proof. The leading term in (4.2) has the sign $\mu(a_N)$ and a positive magnitude. □

Proposition 19.4 (Existence of a finite rounding threshold)**.** *For every $N \geq 2$ there exists a finite $\sigma_N > 1$ such that*

$$\operatorname{nint} \Phi_N(\sigma) = M(N) \tag{19.5}$$

for all real $\sigma \geq \sigma_N$, where nint *denotes the nearest integer.*

Proof. The right-hand side of (4.5) tends to zero because $q_N/N < 1$. It is therefore eventually smaller than $1/2$, and Corollary 4.5 applies. □

Proposition 19.5 (A sufficient explicit threshold)**.** *Let*

$$\lambda_N = \log \frac{N}{q_N} > 0. \tag{19.6}$$

Every $\sigma > 1$ satisfying

$$\sigma \lambda_N > \log(2 h_N N) + \log \frac{1}{\sigma - 1} \tag{19.7}$$

is a valid rounding point. In particular, any $\sigma \geq 2$ satisfying

$$\sigma > \frac{\log(2h_N N)}{\lambda_N} \tag{19.8}$$

is sufficient.

Proof. Take logarithms in (4.9). When $\sigma \geq 2$, $1/(\sigma-1) \leq 1$, which gives the simpler condition. □

19.3. **Recursive recovery of the nonzero tail.**

Definition 19.6. Let

$$a_{N,1} < a_{N,2} < a_{N,3} < \cdots \tag{19.9}$$

be the increasing sequence of integers exceeding N for which $\mu(a_{N,j}) \neq 0$.

Theorem 19.7 (Recursive exponential extraction)**.** *The real-axis function $E_N(\sigma)$ determines the entire signed sequence*

$$\big(a_{N,j}, \mu(a_{N,j})\big)_{j\geq 1} \tag{19.10}$$

recursively. More precisely, once the first $J-1$ terms are known, define

$$E_N^{(J)}(\sigma) = E_N(\sigma) - \frac{\sum_{j<J} \mu(a_{N,j}) a_{N,j}^{-\sigma}}{P_N(\sigma)}. \tag{19.11}$$

Then

$$a_{N,J} = q_N \lim_{\sigma\to\infty} |E_N^{(J)}(\sigma)|^{-1/\sigma}, \tag{19.12}$$

$$\mu(a_{N,J}) = \lim_{\sigma\to\infty} \operatorname{sgn} E_N^{(J)}(\sigma). \tag{19.13}$$

Proof. After subtracting the first $J-1$ tail terms, the numerator is

$$\sum_{j\geq J} \mu(a_{N,j}) a_{N,j}^{-\sigma}. \tag{19.14}$$

Its leading term is the one indexed by J. The proof of Theorem 4.2 applies verbatim. □

Remark 19.8. Theorem 19.7 is an exact uniqueness statement. It does not by itself yield a computational advantage, because evaluating E_N already requires the reciprocal zeta function with sufficient precision and control of exponentially small differences.

20. Additional structure of the positive horizontal defect

20.1. **Even-moment expansion.**

Proposition 20.1 (Power-series expansion in $\log q$)**.** *At a fixed ordinate τ, put*

$$\delta_\rho = \beta - \frac{1}{2} \tag{20.1}$$

for zeros with $\beta > 1/2$. Then

$$D_q(\tau) = \sum_{m=1}^{\infty} \frac{(\log q)^{2m}}{(2m)!} \sum_{\substack{\rho=\beta+i\tau \\ \beta>1/2}} \delta_\rho^{2m}. \tag{20.2}$$

Proof. Expand each $\cosh(\delta_\rho \log q) - 1$ into its absolutely convergent even power series. There are finitely many zeros at a fixed ordinate, counted with multiplicity. □

Corollary 20.2 (Quadratic lower bound)**.** *For every $q > 1$,*

$$D_q(\tau) \geq \frac{(\log q)^2}{2} \sum_{\substack{\rho=\beta+i\tau \\ \beta>1/2}} \left(\beta - \frac{1}{2}\right)^2. \tag{20.3}$$

Proof. Use $\cosh x - 1 \geq x^2/2$. □

Proposition 20.3 (Monotonicity and convexity)**.** *For fixed τ, the function*

$$t \longmapsto D_{e^t}(\tau), \qquad t \geq 0, \tag{20.4}$$

is nonnegative, increasing, and convex. It is identically zero if and only if every zero at ordinate τ lies on the critical line.

Proof. Each summand is $\cosh(\delta t) - 1$ with $\delta > 0$, whose first and second derivatives are nonnegative for $t \geq 0$. □

Corollary 20.4 (One parameter and many parameters)**.** *One fixed $q > 1$ suffices for the logical criterion* (8.10)*. A continuum of values of q additionally encodes all even power sums of the horizontal displacements at each ordinate through* (20.2)*.*

20.2. **Jump formulation.**

Definition 20.5. Define the right-continuous cumulative functions

$$U_q(T) = \sum_{0<\tau\leq T} m(\tau) e^{i\tau \log q}, \tag{20.5}$$

$$V_q(T) = \sum_{0<\tau\leq T} W_q(\tau) e^{i\tau \log q}. \tag{20.6}$$

Proposition 20.6 (Exact jump defect)**.** *At every ordinate τ,*

$$\boxed{\frac{1}{2} e^{-i\tau \log q} \Delta(V_q - U_q)(\tau) = D_q(\tau).} \tag{20.7}$$

Proof. The jump of $V_q - U_q$ at τ is

$$(W_q(\tau) - m(\tau)) e^{i\tau \log q}. \tag{20.8}$$

Apply (8.3). □

21. Regularized explicit formula with explicit prime-power weights

21.1. **Gaussian convolution of a finite interval.**

Definition 21.1. For $\eta > 0$, define

$$g_\eta(t) = \frac{\eta}{\sqrt{\pi}} e^{-\eta^2 t^2}, \tag{21.1}$$

and

$$H_{T,\eta}(z) = \int_0^T g_\eta(z - u)\, \mathrm{d}u. \tag{21.2}$$

The oscillatory test is

$$h_{T,\eta,\omega}(z) = e^{i\omega z} H_{T,\eta}(z). \tag{21.3}$$

Proposition 21.2 (Fourier transform)**.** *With the convention*

$$\widehat{f}(v) = \int_{\mathbb{R}} f(t) e^{-ivt}\,\mathrm{d}t, \tag{21.4}$$

one has

$$\boxed{\widehat{h}_{T,\eta,\omega}(v) = e^{-(v-\omega)^2/(4\eta^2)} \frac{1 - e^{-i(v-\omega)T}}{i(v-\omega)},} \tag{21.5}$$

where the quotient is interpreted continuously as T at $v = \omega$.

Proof. The Fourier transform of g_η is $e^{-v^2/(4\eta^2)}$. Convolution with $\mathbf{1}_{[0,T]}$ multiplies this by $(1 - e^{-ivT})/(iv)$, and multiplication by $e^{i\omega t}$ translates the Fourier variable by ω. □

Proposition 21.3 (Regularized two-branch prime contribution)**.** *In the prime-power side of the Guinand–Weil explicit formula, the test* (21.3) *produces, apart from terms independent of T, the oscillatory sum*

$$\mathcal{P}^{\mathrm{osc}}_{\eta,\omega}(T) = \sum_{n=p^r} c_n^- e^{-(\ell_n-\omega)^2/(4\eta^2)} e^{i(\omega-\ell_n)T} \tag{21.6}$$

$$+ \sum_{n=p^r} c_n^+ e^{-(\ell_n+\omega)^2/(4\eta^2)} e^{i(\omega+\ell_n)T}. \tag{21.7}$$

Proof. Insert $v = \log n$ and $v = -\log n$ into (21.5), multiply by the explicit-formula weight $-\Lambda(n)/(2\pi\sqrt{n})$, and separate the constant and oscillatory parts of $1 - e^{\pm i\alpha T}$. The resulting oscillatory coefficients are (9.2) and (9.3). □

21.2. **A uniform bound for the regularized prime tail.**

Theorem 21.4 (Uniform Gaussian tail)**.** *Assume $Q > e^{2\omega}$ and put $L = \log Q$. If $L > \omega + \eta^2$, then*

$$\sup_{T\in\mathbb{R}} \left| \sum_{\substack{n=p^r \\ n>Q}} \left[c_n^- e^{-(\ell_n-\omega)^2/(4\eta^2)} e^{i(\omega-\ell_n)T} + c_n^+ e^{-(\ell_n+\omega)^2/(4\eta^2)} e^{i(\omega+\ell_n)T} \right] \right|$$

$$\ll_\omega \frac{\eta^2}{L - \omega - \eta^2} \exp\left(\frac{\eta^2}{4} - \frac{(L-\omega-\eta^2)^2}{4\eta^2} \right). \tag{21.8}$$

Proof. For $n > Q > e^{2\omega}$,

$$|\ell_n - \omega| \geq \frac{1}{2}\ell_n, \qquad \ell_n + \omega \geq \ell_n. \tag{21.9}$$

Using $\Lambda(n) \leq \log n$, the absolute value of the tail is bounded by a constant times

$$\sum_{n>Q} n^{-1/2} e^{-(\log n-\omega)^2/(4\eta^2)}. \tag{21.10}$$

Comparison with an integral and the substitution $u = \log x$ yield

$$\int_Q^\infty x^{-1/2} e^{-(\log x-\omega)^2/(4\eta^2)}\,\mathrm{d}x = \int_L^\infty e^{u/2-(u-\omega)^2/(4\eta^2)}\,\mathrm{d}u \tag{21.11}$$

$$= e^{\omega/2+\eta^2/4} \int_L^\infty e^{-(u-\omega-\eta^2)^2/(4\eta^2)}\,\mathrm{d}u. \tag{21.12}$$

Apply the standard Gaussian tail bound

$$\int_r^\infty e^{-v^2/(4\eta^2)}\,\mathrm{d}v \leq \frac{2\eta^2}{r} e^{-r^2/(4\eta^2)}. \tag{21.13}$$

□

Corollary 21.5 (One explicit joint choice)**.** *If*

$$\log Q = \omega + 3\eta^2, \tag{21.14}$$

then the right-hand side of (21.8) *is*

$$O_\omega(e^{-3\eta^2/4}). \tag{21.15}$$

Warning 21.6. Theorem 21.4 controls the prime side of a regularized explicit formula. It does not identify the complete-zero spectral side with the unweighted ordinate walk. That identification is exactly where the positive measure (8.9) appears.

22. Mean-square statements and their exact logical strength

22.1. **Finite Dirichlet-polynomial tails in Lebesgue mean.**

Proposition 22.1 (Finite-tail mean-value estimate)**.** *Let*

$$E_{Q,R}(T) = \sum_{\substack{Q<n\le R\\ n=p^r}} \left(c_n^- n^{-iT} + c_n^+ n^{iT}\right). \tag{22.1}$$

Then

$$\frac{1}{X}\int_0^X |E_{Q,R}(T)|^2\,\mathrm{d}T \ll_\omega \left(1+\frac{R}{X}\right)\left[\log\frac{\log R}{\log Q} + \frac{1}{\log Q} + Q^{-1/2}\right] \tag{22.2}$$

for $e^{2\omega} < Q < R$.

Proof. Apply the Montgomery–Vaughan mean-value theorem for Dirichlet polynomials [3] to the two branches separately. For a prime power $n = p^r$,

$$|c_n^-|^2 + |c_n^+|^2 \ll_\omega \frac{\Lambda(n)^2}{n(\log n)^2} = \frac{1}{r^2 p^r}. \tag{22.3}$$

The prime terms contribute

$$\sum_{Q<p\le R}\frac{1}{p} = \log\frac{\log R}{\log Q} + O\left(\frac{1}{\log Q}\right), \tag{22.4}$$

while the proper powers contribute $O(Q^{-1/2})$. □

Corollary 22.2 (A finite logarithmic tail can be exhausted)**.** *Let*

$$R(X) = X^{1/2}, \qquad Q(X) = R(X)^{1-1/\sqrt{\log\log X}}. \tag{22.5}$$

Then

$$\frac{1}{X}\int_0^X |E_{Q(X),R(X)}(T)|^2\,\mathrm{d}T \longrightarrow 0. \tag{22.6}$$

Warning 22.3. Corollary 22.2 concerns a finite tail in Lebesgue measure. The numerical variance is sampled on zeta ordinates and contains the projected kernel (15.8); the two norms are not interchangeable.

22.2. **Normalized global convergence does not detect fixed atoms.**

Proposition 22.4 (Counterexample to global normalized sufficiency)**.** *There exist functions F_X such that*

$$\frac{1}{X}\int_0^X \left|F_X(T)-\frac{1}{2}\right|^2 \,\mathrm{d}T \longrightarrow 0, \tag{22.7}$$

but whose distributional derivatives retain a nonzero fixed atomic part.

Proof. Take

$$F_X(T)=\frac{1}{2}+\mathbf{1}_{[0,1]}(T). \tag{22.8}$$

Then the normalized square integral is $1/X$, while

$$\frac{\mathrm{d}}{\mathrm{d}T}\mathbf{1}_{[0,1]}=\delta_0-\delta_1 \tag{22.9}$$

in distributions. □

Corollary 22.5. *Any route from energy exhaustion to the vanishing of the positive atomic measure* (8.9) *must use a local topology, atomwise convergence, or distributional convergence against compactly supported test functions.*

23. Weighted prime-power profiles

Definition 23.1. Let $W:[0,\infty)\to\mathbb{C}$ be bounded and compactly supported. Define

$$G_{Q,W}(T)=\sum_{n=p^r} W\left(\log\frac{n}{Q}\right) g_n(T). \tag{23.1}$$

Theorem 23.2 (Weighted Bohr energy)**.** *As $Q\to\infty$,*

$$\lim_{X\to\infty}\frac{1}{X}\int_0^X |G_{Q,W}(T)|^2\,\mathrm{d}T=\frac{1}{2\pi^2\log Q}\int_0^\infty |W(u)|^2\,\mathrm{d}u+o\left(\frac{1}{\log Q}\right). \tag{23.2}$$

Proof. Bohr orthogonality removes all unequal frequencies. The prime contribution is

$$\frac{1}{2\pi^2}\sum_p \frac{1}{p}\left|W\left(\log\frac{p}{Q}\right)\right|^2+O_\omega\left(\frac{1}{(\log Q)^2}\right). \tag{23.3}$$

Partial summation with the prime number theorem and the substitution $p=Qe^u$ gives the stated integral. Proper prime powers are lower order. □

Corollary 23.3 (Interpretation of the constant $3/4$)**.** *A weighted Bohr model has leading energy $3/(4\pi^2\log Q)$ precisely when*

$$\int_0^\infty |W(u)|^2\,\mathrm{d}u=\frac{3}{2}. \tag{23.4}$$

The constant determines a quadratic profile mass, not a unique hard support.

24. Detailed projected Gram algebra

24.1. **Four branch expansion.**

Proposition 24.1 (Full unprojected quadratic expansion)**.** *For a finite set $\mathcal{N}$ of prime powers, put*

$$G_{\mathcal{N}}(\gamma)=\sum_{n\in\mathcal{N}}\left(c_n^- n^{-i\gamma}+c_n^+ n^{i\gamma}\right). \tag{24.1}$$

Then

$$\|G_{\mathcal{N}}\|_H^2 = \sum_{m,n\in\mathcal{N}} \overline{c_m^-} c_n^- M_H(m/n) \tag{24.2}$$

$$+ \sum_{m,n\in\mathcal{N}} \overline{c_m^-} c_n^+ M_H(mn) \tag{24.3}$$

$$+ \sum_{m,n\in\mathcal{N}} \overline{c_m^+} c_n^- \overline{M_H(mn)} \tag{24.4}$$

$$+ \sum_{m,n\in\mathcal{N}} \overline{c_m^+} c_n^+ M_H(n/m). \tag{24.5}$$

Proof. Expand the square and apply the definition (12.4) to each of the four sign combinations. □

24.2. **Prime-tower blocks.**

Proposition 24.2 (Same-tower resonance closure)**.** *For $m = p^r$ and $n = p^s$, every argument of M_H in* (24.2)–(24.5) *is one of*

$$p^{s-r}, \qquad p^{r-s}, \qquad p^{r+s}, \qquad p^{-(r+s)}. \tag{24.6}$$

Thus an entire two-branch prime tower is an arithmetically resonant Gram block; it is not reduced to the scalar diagonal $r = s$.

Proof. Substitute $m = p^r$ and $n = p^s$ into the four arguments in Proposition 24.1. □

Proposition 24.3 (Projection correction)**.** *Let V be the matrix whose columns are the sampled prime-power modes, and let B be the two-column matrix with columns* 1 *and $e^{i\omega\gamma}$. Then the residual Gram matrix is*

$$\widetilde{K}_H = \frac{1}{N(H)} V^*V - \frac{1}{N(H)^2} V^*B \left(\frac{1}{N(H)} B^*B \right)^{-1} B^*V. \tag{24.7}$$

In particular, even the norm of a single frequency is altered by the constant–carrier projection.

Proof. The orthogonal projection onto the complement of the column space of B is

$$I - B(B^*B)^{-1}B^*. \tag{24.8}$$

Insert it between V^* and V and normalize. □

25. Analytic targets isolated by the exact identities

Conjecture 25.1 (Collective carrier law)**.** *In a specified joint range $Q = Q(H) \to \infty$,*

$$\Im \widehat{c}_{Q,H} = \pi \Delta_\omega(Q) + o\left(\Delta_\omega(Q)\right). \tag{25.1}$$

By Corollary 12.6, this is equivalent to a weighted collective cancellation in the exact kernel $M_H(n)$.

Conjecture 25.2 (Projected variance law)**.** *For a suitable joint range,*

$$V(Q,H) = \frac{3}{4} \Delta_\omega(Q) + o\left(\Delta_\omega(Q)\right). \tag{25.2}$$

The left-hand side is the projected discrete quadratic form (15.12)*; no ordinary Parseval diagonal is assumed.*

Question 25.3 (Tower and cross-tower separation)**.** Can one prove, uniformly in a growing cutoff, asymptotic formulas for

$$\sum_p a_p^* \widetilde{K}_{p,p;H} a_p \tag{25.3}$$

and

$$\sum_{p\neq q} a_p^* \widetilde{K}_{p,q;H} a_q \tag{25.4}$$

separately? This is the exact decomposition required before any assertion about a "diagonal" can be made.

26. Open problems and falsifiable next computations

Question 26.1 (Increment audit)**.** For every prime power n in a growing range, compute the exact ratio

$$\mathcal{I}_H(n) = \frac{\Re M_H(n)}{\pi v_n} \tag{26.1}$$

and compare it with

$$\frac{\log n}{\log(H/2\pi)}. \tag{26.2}$$

This tests the local Landau mechanism without fitting a global law.

Question 26.2 (Height–cutoff matrix)**.** Repeat the complete computation for a two-dimensional grid of (H, Q), including regimes

$$Q = H^\theta \tag{26.3}$$

with several fixed θ and a near-height regime $Q = H^{1-o(1)}$. Record raw and joint coefficients separately.

Question 26.3 (Projected tower blocks)**.** Compute

$$a_p^* \widetilde{K}_{p,p;H} a_p \tag{26.4}$$

for every prime tower and compare its cumulative sum with the cross-tower term in (15.13).

Question 26.4 (Local test-function convergence)**.** For compactly supported smooth φ, determine whether the regularized residual satisfies

$$\int \left(F_{Q,H}(T) - \frac{1}{2} \right) \varphi(T)\, \mathrm{d}T \longrightarrow 0 \tag{26.5}$$

under a joint limiting regime, while the prime-side derivative is identified with the complete spectral measure.

Part 3. Numerical support

27. Computational object and exact program identities

27.1. Input data and sampled height.

Experiment 27.1 (Zero data)**.** The full-data run used 2,001,052 positive ordinates from `zeross.bin`, with

$$\gamma_1 = 14.134725142, \qquad \gamma_{2{,}001{,}052} = 1{,}132{,}490.6587144111. \tag{27.1}$$

The phase parameter was

$$\omega = 1, \qquad q = e. \tag{27.2}$$

All ordinates were used; the stride was one.

27.2. **Exact definition of the program.**

Definition 27.2 (Program variables)**.** At each sampled ordinate, the code forms

$$Y(\gamma_k) = S_q(\gamma_k) - A_{\mathrm{smooth}}(\gamma_k), \tag{27.3}$$

and

$$R_Q(\gamma_k) = Y(\gamma_k) - P_Q^-(\gamma_k) - P_Q^+(\gamma_k), \tag{27.4}$$

where

$$P_Q^-(\gamma) = \sum_{\substack{n=p^r \\ n\le Q}} c_n^- e^{i(\omega-\ell_n)\gamma}, \tag{27.5}$$

$$P_Q^+(\gamma) = \sum_{\substack{n=p^r \\ n\le Q}} c_n^+ e^{i(\omega+\ell_n)\gamma}. \tag{27.6}$$

The raw carrier statistic is

$$\widehat{c}_{Q,H} = \frac{1}{N(H)} \sum_{\gamma_k \le H} e^{-i\omega\gamma_k} R_Q(\gamma_k). \tag{27.7}$$

Proposition 27.3 (Identity evaluated by the raw program)**.** *The program variables satisfy exactly*

$$\widehat{c}_{Q,H} = \widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} \left(c_n^- \overline{M_H(n)} + c_n^+ M_H(n) \right). \tag{27.8}$$

Its imaginary part is

$$\Im\widehat{c}_{Q,H} = \Im\widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} u_n \Re M_H(n). \tag{27.9}$$

Proof. This is Theorems 12.2 and 12.3 applied to the arrays in Definition 27.2. □

27.3. **Joint constant–carrier fit used for the reported values.**

Definition 27.4. For every cutoff, the code fits

$$R_Q(\gamma_k) = \alpha_{Q,H} + c_{Q,H} e^{i\omega\gamma_k} + E_{Q,H}(\gamma_k) \tag{27.10}$$

by least squares in the sampled inner product (15.1). The reported variance is

$$V(Q,H) = \frac{1}{N(H)} \sum_{\gamma_k \le H} |E_{Q,H}(\gamma_k)|^2. \tag{27.11}$$

Proposition 27.5 (Closed form of the reported carrier)**.** *Let*

$$\mu_H = M_H(e^{\omega}). \tag{27.12}$$

Then

$$\boxed{c_{Q,H} = \frac{\langle e^{i\omega\gamma}, R_Q\rangle_H - \overline{\mu_H}\langle 1, R_Q\rangle_H}{1 - |\mu_H|^2}.} \tag{27.13}$$

The fitted constant is

$$\alpha_{Q,H} = \frac{\langle 1, R_Q\rangle_H - \mu_H\langle e^{i\omega\gamma}, R_Q\rangle_H}{1 - |\mu_H|^2}. \tag{27.14}$$

Proof. Solve the two normal equations for the basis $\{1, e^{i\omega\gamma}\}$. □

27.4. **Prime-power generation and exact coefficients.**

Experiment 27.6 (Prime powers)**.** The cutoffs and counts were

Q	number of prime powers $n = p^r \le Q$
0	0
3	2
10	7
100	35
1000	193
10000	1280

Every coefficient was generated from (9.2)–(9.3); no frequency or amplitude was fitted individually.

28. Radial geometry and low prime-power resonances

28.1. **Direct radial audit.**

Experiment 28.1 (Original walk and theoretical subtractions)**.** After subtraction of the empirical complex center, the radial statistics were

case	mean radius	radial standard deviation	smoothed radial modes
original	1.9203104476	0.7265477726	1.35731059, 2.49789956
after $n = 3$	1.8079247921	0.3083120307	1.68017926
after $n = 2, 3$	1.7884459085	0.2214048337	1.81300495

The theoretical $n = 3$ coefficients were

$$c_3^- = -1.02370050729i, \qquad c_3^+ = 0.0481029537851i. \tag{28.1}$$

Subtracting $n = 3$ reduced the radial variance by approximately 81.99%; subtracting $n = 2, 3$ reduced it by approximately 90.71%. The finite conclusion is that the observed global double radial mode is dominated by the theoretical $n = 3$ oscillator.

28.2. **Geometric interpretation.**

Experiment 28.2 (Anisotropy)**.** At $Q = 10000$, the centered residual anisotropy was

$$0.981212. \tag{28.2}$$

This is consistent with the exact single-mode limit (10.14): after low frequencies are removed, large-n modes are strongly elongated in one direction.

29. Cutoff sweep and carrier stabilization

29.1. **Global cutoff table.**

Experiment 29.1 (Full-data cutoff sweep)**.** The jointly fitted carrier and residual energy were

Q	terms	$\Re c_{Q,H}$	$\Im c_{Q,H}$	$V(Q,H)$	$\sqrt{V(Q,H)}$
0	0	0.4999989123	0.1381673175	1.2420277147	1.1144630
3	2	0.4999998679	0.0956865241	0.1339324657	0.3659678
10	7	0.4999999862	0.0755224108	0.0596099186	0.2441514
100	35	0.5000000416	0.0545919879	0.0220396104	0.1484574
1000	193	0.5000000755	0.0426449212	0.0112822119	0.1062178
10000	1280	0.4999998710	0.0343125459	0.0082559398	0.0908622

At the largest cutoff the finite variance reduction relative to $Q = 0$ was

$$1 - \frac{0.0082559398}{1.2420277147} = 0.99335\ldots. \tag{29.1}$$

29.2. **Normalized $1/\log Q$ comparisons.**

Experiment 29.2 (Two normalized quantities)**.** For $Q \geq 3$, the finite normalized values were

Q	$\pi \log Q\, \Im c_{Q,H}$	$\frac{4\pi^2}{3} \log Q\, V(Q,H)$
3	0.33025	1.93628
10	0.54631	1.80623
100	0.78981	1.33564
1000	0.92545	1.02558
10000	0.99284	1.00065

At $Q = 10000$,

$$\frac{1}{\pi \log Q} = 0.0345601\ldots, \tag{29.2}$$

$$\frac{3}{4\pi^2 \log Q} = 0.00825061\ldots. \tag{29.3}$$

The corresponding measured values were 0.0343125459 and 0.0082559398. These are finite coincidences to be interpreted through the exact defect identities, not proofs of asymptotic laws.

30. Height dependence

30.1. **Twenty-four block audit.**

Experiment 30.1 (Block ranges)**.** For $Q = 10000$ and 24 consecutive height blocks,

$$\Re\, \mathrm{mean}_{\mathrm{block}}\, F_Q \in [0.499994991361, 0.500003656384], \tag{30.1}$$

$$\Im\, \mathrm{mean}_{\mathrm{block}}\, F_Q \in [0.033825249952, 0.035105891286], \tag{30.2}$$

$$V_{\mathrm{block}} \in [0.0054744557, 0.0089851833]. \tag{30.3}$$

The standard deviation of the block means was

$$0.0003812589. \tag{30.4}$$

The carrier is stable across height at this scale, while the block variance has a visible drift. Consequently the global factor 3/4 cannot yet be regarded as height-uniform.

30.2. **Termwise increment scale.**

Experiment 30.2 (Present range of the Landau ratio)**.** For

$$H = 1.13249 \times 10^6, \qquad Q = 10^4, \tag{30.5}$$

one has approximately

$$\frac{\log Q}{\log(H/2\pi)} \approx 0.76. \tag{30.6}$$

The global ratio

$$\frac{\Im c_{Q,H}}{\pi \Delta_1(Q)} \approx 0.986 \tag{30.7}$$

is therefore not explained by independent termwise transport at the leading Landau scale. The exact increment statistic (12.22) is the direct falsifiable test.

31. Branch decomposition and omitted frequencies

31.1. **Branch energies.**

Experiment 31.1 (Minus and plus branches)**.** At $Q = 10000$,

$$\|P_Q^-\|^2 = 1.2433344613, \tag{31.1}$$

$$\|P_Q^+\|^2 = 0.0309962015, \tag{31.2}$$

$$2\Re\langle P_Q^-, P_Q^+\rangle = -0.0005963448. \tag{31.3}$$

The minus branch dominates, and direct plus–minus energy cancellation is small.

31.2. **Omitted prime-power shell.**

Experiment 31.2 (Exact-frequency omitted fit)**.** For the 56 omitted prime powers immediately above $Q = 10000$, the median amplitude-to-null ratios were

$$3.35349 \quad \text{and} \quad 2.67559 \tag{31.4}$$

for the minus and plus branches. The absolute correlation between the jointly fitted coefficient vector and the theoretical coefficient vector was

$$0.916474. \tag{31.5}$$

The narrow shell explained

$$0.0144446 \tag{31.6}$$

of the remaining residual energy. Thus the next prime-power frequencies are present, but this narrow shell does not account for the full residual.

32. Signed invariant and finite-area comparisons

32.1. **Direct evaluation of $\Delta_1(Q)$.**

Experiment 32.1 (Common invariant)**.** At $Q = 10000$,

$$\Delta_1(10000) \approx 0.01108. \tag{32.1}$$

Hence

$$\pi\Delta_1(10000) \approx 0.03481, \qquad \frac{3}{4}\Delta_1(10000) \approx 0.00831. \tag{32.2}$$

The same exact signed invariant therefore reproduces both observed scales to finite accuracy. The analytic claims that the centroid and projected variance equal these multiples remain Conjectures 25.1 and 25.2.

32.2. **Coefficient prefix sums.**

Experiment 32.2 (Prefix sums)**.** The coefficient combinations

$$A_Q = \sum_{n\le Q}(c_n^- + c_n^+), \qquad B_Q = \sum_{n\le Q}(c_n^- - c_n^+) \tag{32.3}$$

were, at $Q = 10000$,

$$A_Q = -2.5552137617i, \qquad B_Q = -10.6932626383i. \tag{32.4}$$

They do not account for the real carrier value $1/2$.

33. Finite-part computations for Φ_N

33.1. **Direct algebraic verification.**

Experiment 33.1 (Finite-part identity)**.** A numerical comparison between the direct tail expression and the value reconstructed from Proposition 6.3 gave

$$|R_{\text{from FP}} - R_{\text{direct}}| = 4.043 \times 10^{-10}. \tag{33.1}$$

This verifies the implemented identity at the working precision.

33.2. **Scale scan.**

Experiment 33.2 (Normalized finite-part differences)**.** For

$$N = 2000, 4000, 8000, 16000, 32000, 64000, \tag{33.2}$$

the successive magnitudes

$$|\mathcal{F}_{2N}(\rho) - \mathcal{F}_N(\rho)| \tag{33.3}$$

were approximately

$$1.12, \quad 0.75, \quad 0.80, \quad 0.75, \quad 0.44. \tag{33.4}$$

No monotone cancellation law is established by these six scales.

34. Numerical tests dictated by the analytic identities

34.1. **Prime-power increment audit.**

Experiment 34.1 (Required output)**.** For every prime power n in the selected range, compute and store

$$\mathcal{I}_H(n) = \frac{\Re M_H(n)}{\pi v_n}, \tag{34.1}$$

together with

$$\theta_H(n) = \frac{\log n}{\log(H/2\pi)}. \tag{34.2}$$

The comparison is pointwise and requires no global fitting.

34.2. **Two-parameter height–cutoff scan.**

Experiment 34.2 (Required grid)**.** For several heights H_j and exponents θ, set

$$Q_{j,\theta} = H_j^{\theta} \tag{34.3}$$

and record separately:

$$\widehat{c}_{Q,H}, \quad c_{Q,H}, \quad V(Q,H), \quad \Delta_\omega(Q), \quad \sum_p a_p^* \widetilde{K}_{p,p;H} a_p, \quad \sum_{p \neq q} a_p^* \widetilde{K}_{p,q;H} a_q. \tag{34.4}$$

This directly separates raw carrier, projected carrier, tower blocks, and cross-tower interactions.

35. Complete MATLAB implementation

35.1. **Self-contained implementation.**

Remark 35.1. The following program is the complete implementation used for the prime-power audit. It reads only the original binary ordinate file, generates all prime powers internally, retains all figures, performs the joint constant–carrier fit, and exports every table used in the preceding numerical sections.

Listing 1. Complete MATLAB implementation of the prime-power phase audit.

```
%% ZETA_PRIME_POWER_ANALYTIC_AUDIT
%
% Script autocontenido para los seis tests de la descomposicion
%
%   S_q(T) = sum_{gamma_j <= T} exp(i*gamma_j*log(q))
%
% No usa ningun .mat, .csv ni resultado anterior.
% El unico dato externo necesario es el fichero original de ordenadas:
%
%   zeross.bin
%
% Si no esta en la carpeta actual, MATLAB abre un selector para localizarlo.
%
% No resta 0.5*exp(i*T). Ajusta conjuntamente una constante aditiva y
% el coeficiente de exp(i*log(q)*T). La constante se elimina antes de
% demodular.
% Todos los resumenes principales se muestran con fprintf/disp.
% No se cierra ninguna figura.
%
% TESTS
% -----
% 1. Separacion P_Q^- y P_Q^+ y termino cruzado.
% 2. Identidades de c_n^-, c_n^+ y sumas A_Q, B_Q.
% 3. Busqueda de telescopado dentro de p,p^2,...,p^R.
% 4. Funcionales positivos y matrices de Gram por bloques.
% 5. Ecuacion funcional del residuo demodulado F_Q.
% 6. Unicidad: bloques, anisotropia, gaps y frecuencias omitidas.
%
% Requiere una version de MATLAB que permita funciones locales al final
% de un script (R2016b o posterior).

clear;
clc;
format long g;

%% ========================================================================
% CONFIGURACION
% ========================================================================

zeroBinaryFile = 'zeross.bin';

% q=e reproduce exp(i*gamma).
qPhase = exp(1);
omega = log(qPhase);

% stride=10: usa todos los ceros para construir S_q, pero analiza 1 de cada 10.
% stride=1: analiza los 2,001,052 puntos completos, mucho mas lento para Q=10000.
stride = 1;

Qlist = [0, 3, 10, 100, 1000, 10000];
Qlist = unique(sort(Qlist(:).'));
Qmax = max(Qlist);

```

```
numHeightBlocks = 24;
numGramBlocks = 20;
maxGramPoints = 60000;

omittedTestWidth = 500;
maxSpectralPoints = 50000;
numFrequencyGridPoints = 401;

gapMedianWindow = 1001;
resonanceTolerance = 1e-12;
minimumVariance = 1e-30;

% Si se pone true, guarda tablas y MAT al final. No son necesarios para correr.
saveResults = false;
outputDirectory = 'ZETA_PRIME_POWER_STRUCTURE_RESULTS';

%% ========================================================================
% LOCALIZAR Y LEER zeross.bin
% ========================================================================

if ~isfile(zeroBinaryFile)
    [selectedFile,selectedPath] = uigetfile( ...
        {'*.bin','Binary doubles (*.bin)';'*.*','Todos los archivos'}, ...
        'Selecciona zeross.bin');

    if isequal(selectedFile,0)
        error(['No se encontro zeross.bin y no se selecciono ningun archivo. ' ...
            'Este es el unico fichero de datos necesario.']);
    end

    zeroBinaryFile = fullfile(selectedPath,selectedFile);
end

fileID = fopen(zeroBinaryFile,'rb');
if fileID == -1
    error('No se pudo abrir %s.',zeroBinaryFile);
end

cleanupObject = onCleanup(@() fclose(fileID));
gammaAll = fread(fileID,Inf,'double=>double');
clear cleanupObject;

gammaAll = gammaAll(:);
gammaAll = gammaAll(isfinite(gammaAll) & gammaAll>0);

if numel(gammaAll)<2
    error('El fichero no contiene suficientes ordenadas validas.');
end

if any(diff(gammaAll)<=0)
    error('Las ordenadas no estan estrictamente ordenadas.');
end

Nfull = numel(gammaAll);

fprintf('\n============================================================\n');
fprintf('ZETA PRIME-POWER STRUCTURE: ANALYTIC AUDIT\n');
fprintf('============================================================\n');
fprintf('Archivo de ceros       : %s\n',zeroBinaryFile);
fprintf('Ceros disponibles      : %d\n',Nfull);
fprintf('q                      : %.17g\n',qPhase);
fprintf('omega=log(q)           : %.17g\n',omega);
fprintf('stride                 : %d\n',stride);
fprintf('Qlist                  :');
fprintf(' %g',Qlist);
```

```
fprintf('\n');

%% ========================================================================
% CONSTRUIR S_q CON TODOS LOS CEROS; MUESTREAR DESPUES
% ========================================================================

carrierAll = exp(1i*omega*gammaAll);
SphaseAll = cumsum(carrierAll);

% Parte suave analitica, salvo constante aditiva:
%
% int exp(i*omega*T)/(2*pi)*log(T/(2*pi)) dT
%
% = [exp(i omega T)log(T/2pi)-Ei(i omega T)]/(2*pi*i*omega).
%
% MATLAB: Ei(z) = -expint(-z), salvo la constante de rama, que solo
% modifica una constante aditiva y se absorbe en los ajustes.
Ei_iomegaT_all = -expint(-1i*omega*gammaAll);
AsmoothAll = (carrierAll.*log(gammaAll/(2*pi))-Ei_iomegaT_all) ...
    /(2*pi*1i*omega);

indices = (1:stride:Nfull).';
gamma = gammaAll(indices);
carrier = carrierAll(indices);
Sphase = SphaseAll(indices);
Asmooth = AsmoothAll(indices);

gapBeforeAll = [NaN;diff(gammaAll)];
gapLocalMedianAll = movmedian(gapBeforeAll,gapMedianWindow, ...
    'omitmissing','Endpoints','shrink');
gapResidualAll = gapBeforeAll-gapLocalMedianAll;

gapBefore = gapBeforeAll(indices);
gapResidual = gapResidualAll(indices);

clear carrierAll SphaseAll AsmoothAll Ei_iomegaT_all;
clear gapBeforeAll gapLocalMedianAll gapResidualAll;

Y = Sphase-Asmooth;
N = numel(gamma);

fprintf('Puntos analizados       : %d\n',N);
fprintf('gamma inicial           : %.15g\n',gamma(1));
fprintf('gamma final             : %.15g\n',gamma(end));

%% ========================================================================
% GENERAR TODAS LAS POTENCIAS PRIMAS <= Qmax
% ========================================================================

[qVec,basePrimeVec,powerExponentVec,lambdaVec] = makePrimePowers(Qmax);
numTerms = numel(qVec);

ellVec = log(qVec);
deltaMinusVec = omega-ellVec;
deltaPlusVec = omega+ellVec;

resonantIndex = find(abs(deltaMinusVec)<=resonanceTolerance,1);
if ~isempty(resonantIndex)
    error(['qPhase es resonante con n=%d. Esta version no oculta la ' ...
        'singularidad: debe introducirse el termino resonante regularizado.'], ...
        qVec(resonantIndex));
end

cMinusVec = 1i*lambdaVec ./ ...
    (2*pi*sqrt(qVec).*deltaMinusVec);
```

```
184
185 cPlusVec = 1i*lambdaVec ./ ...
186     (2*pi*sqrt(qVec).*deltaPlusVec);
187
188 %% ========================================================================
189 % TEST 2: IDENTIDADES ALGEBRAICAS DE LOS COEFICIENTES
190 % ========================================================================
191
192 sumIdentityRHS = 1i*lambdaVec.*omega ./ ...
193     (pi*sqrt(qVec).*(omega^2-ellVec.^2));
194
195 differenceIdentityRHS = 1i*lambdaVec.*ellVec ./ ...
196     (pi*sqrt(qVec).*(omega^2-ellVec.^2));
197
198 ratioIdentityRHS = (omega+ellVec)./(omega-ellVec);
199
200 sumIdentityError = abs((cMinusVec+cPlusVec)-sumIdentityRHS);
201 differenceIdentityError = abs((cMinusVec-cPlusVec)-differenceIdentityRHS);
202 ratioIdentityError = abs((cMinusVec./cPlusVec)-ratioIdentityRHS);
203
204 AQ = cumsum(cMinusVec+cPlusVec);
205 BQ = cumsum(cMinusVec-cPlusVec);
206
207 fprintf('\n=============================================================\n');
208 fprintf('TEST 2: IDENTIDADES DE COEFICIENTES\n');
209 fprintf('=============================================================\n');
210 fprintf('Potencias primas <=Qmax                 : %d\n',numTerms);
211 fprintf('max |(c^-+c^+)-formula|                : %.6e\n', ...
212     max(sumIdentityError));
213 fprintf('max |(c^--c^+)-formula|                : %.6e\n', ...
214     max(differenceIdentityError));
215 fprintf('max |c^-/c^+-formula|                  : %.6e\n', ...
216     max(ratioIdentityError));
217
218 coefficientCutoffRows = cell(numel(Qlist),9);
219
220 for jj = 1:numel(Qlist)
221     currentQ = Qlist(jj);
222     lastIndex = find(qVec<=currentQ,1,'last');
223
224     if isempty(lastIndex)
225         currentA = 0;
226         currentB = 0;
227         numberTerms = 0;
228     else
229         currentA = AQ(lastIndex);
230         currentB = BQ(lastIndex);
231         numberTerms = lastIndex;
232     end
233
234     coefficientCutoffRows(jj,:) = { ...
235         currentQ,numberTerms, ...
236         real(currentA),imag(currentA),abs(currentA-0.5), ...
237         real(currentB),imag(currentB),abs(currentB-0.5), ...
238         abs(currentB+0.5)};
239 end
240
241 coefficientCutoffDisplay = cell2table(coefficientCutoffRows, ...
242     'VariableNames',{ ...
243     'Q','num_terms','Re_AQ','Im_AQ','abs_AQ_minus_half', ...
244     'Re_BQ','Im_BQ','abs_BQ_minus_half','abs_BQ_plus_half'});
245
246 disp(coefficientCutoffDisplay);
247
248
```

```
%% ======================================================================
% AUDITORIA RADIAL DIRECTA: ORIGINAL, n=3, n=2+3
% ======================================================================

fprintf('\n============================================================\n');
fprintf('AUDITORIA RADIAL DIRECTA DE LA BIMODALIDAD\n');
fprintf('============================================================\n');

numRadialBins = 220;
radialSmoothWidth = 9;

index3 = find(qVec==3,1);
index23 = find(qVec<=3);

if isempty(index3)
    error('No se encontro n=3 entre las potencias primas.');
end

ell3 = ellVec(index3);
P3minus = cMinusVec(index3).*carrier.*exp(-1i*ell3*gamma);
P3plus = cPlusVec(index3).*carrier.*exp(1i*ell3*gamma);
P3 = P3minus+P3plus;

P23 = complex(zeros(N,1));
for auditIndex = index23(:).'
    ellAudit = ellVec(auditIndex);
    P23 = P23 ...
        +cMinusVec(auditIndex).*carrier.*exp(-1i*ellAudit*gamma) ...
        +cPlusVec(auditIndex).*carrier.*exp(1i*ellAudit*gamma);
end

[radialOriginal,radialCentersOriginal,radialDensityOriginal, ...
    radialPeaksOriginal] = radialDistributionSummary( ...
    Sphase,numRadialBins,radialSmoothWidth);

[radialAfter3,radialCentersAfter3,radialDensityAfter3, ...
    radialPeaksAfter3] = radialDistributionSummary( ...
    Sphase-P3,numRadialBins,radialSmoothWidth);

[radialAfter23,radialCentersAfter23,radialDensityAfter23, ...
    radialPeaksAfter23] = radialDistributionSummary( ...
    Sphase-P23,numRadialBins,radialSmoothWidth);

radialAuditTable = table( ...
    ["original";"after_n3";"after_n2_n3"], ...
    [mean(radialOriginal);mean(radialAfter3);mean(radialAfter23)], ...
    [std(radialOriginal);std(radialAfter3);std(radialAfter23)], ...
    [numel(radialPeaksOriginal);numel(radialPeaksAfter3); ...
        numel(radialPeaksAfter23)], ...
    [joinNumericVector(radialPeaksOriginal); ...
        joinNumericVector(radialPeaksAfter3); ...
        joinNumericVector(radialPeaksAfter23)], ...
    'VariableNames',{'case_name','mean_radius','std_radius', ...
    'num_detected_peaks','detected_peak_radii'});

disp(radialAuditTable);

fprintf('Coeficiente teorico total n=3:\n');
fprintf('  c3^- = %.12g %+.12gi\n',real(cMinusVec(index3)), ...
    imag(cMinusVec(index3)));
fprintf('  c3^+ = %.12g %+.12gi\n',real(cPlusVec(index3)), ...
    imag(cPlusVec(index3)));

figure('Name','Auditoria radial: original y sustracciones pequenas', ...
    'Color','w');
```

```
plot(radialCentersOriginal,radialDensityOriginal,'LineWidth',1.3);
hold on;
plot(radialCentersAfter3,radialDensityAfter3,'LineWidth',1.3);
plot(radialCentersAfter23,radialDensityAfter23,'LineWidth',1.3);
grid on;
xlabel('Radio respecto del centro medio');
ylabel('Densidad suavizada');
legend('Original','Despues de n=3','Despues de n=2,3', ...
    'Location','best');
title('Prueba directa de la bimodalidad radial');

%% ========================================================================
% PREPARAR MATRIZ DE GRAM POR BLOQUES LOGARITMICOS DE n
% ========================================================================

gramIndices = unique(round(linspace(1,N,min(N,maxGramPoints)))).';
numGramPoints = numel(gramIndices);

logEdges = linspace(log(2),log(Qmax)+100*eps(log(Qmax)), ...
    numGramBlocks+1);

gramBlockOfTerm = discretize(ellVec,logEdges);
gramBlockOfTerm(isnan(gramBlockOfTerm)) = numGramBlocks;

gramMinus = complex(zeros(numGramPoints,numGramBlocks,'single'));
gramPlus = complex(zeros(numGramPoints,numGramBlocks,'single'));

%% ========================================================================
% BUCLE PRINCIPAL: TESTS 1, 4, 5 Y PARTE DE 6
% ========================================================================

Pminus = complex(zeros(N,1));
Pplus = complex(zeros(N,1));

nextTerm = 1;
maxIncrementIdentityError = 0;
maxConstantCancellationIdentityError = 0;

summaryRows = cell(numel(Qlist),43);
positivityRows = cell(numel(Qlist),17);

blockRows = cell(numel(Qlist)*numHeightBlocks,16);
blockRow = 0;

[constantY,cY] = fitConstantAndCarrier(Y,carrier);

RbothAtQmax = [];
FAtQmax = [];
EAtQmax = [];
PminusAtQmax = [];
PplusAtQmax = [];
cBothAtQmax = NaN;

for qi = 1:numel(Qlist)

    Q = Qlist(qi);

    while nextTerm<=numTerms && qVec(nextTerm)<=Q

        ell = ellVec(nextTerm);
        z = exp(1i*ell*gamma);

        termMinus = cMinusVec(nextTerm).*carrier.*conj(z);
        termPlus = cPlusVec(nextTerm).*carrier.*z;

```

```
        oldCombinedResidual = Y-Pminus-Pplus;

        Pminus = Pminus+termMinus;
        Pplus = Pplus+termPlus;

        newCombinedResidual = Y-Pminus-Pplus;

        incrementError = max(abs( ...
            (newCombinedResidual-oldCombinedResidual) ...
            +termMinus+termPlus));

        maxIncrementIdentityError = max( ...
            maxIncrementIdentityError,incrementError);

        gramBlock = gramBlockOfTerm(nextTerm);

        gramMinus(:,gramBlock) = gramMinus(:,gramBlock) ...
            +single(termMinus(gramIndices));

        gramPlus(:,gramBlock) = gramPlus(:,gramBlock) ...
            +single(termPlus(gramIndices));

        nextTerm = nextTerm+1;
    end

    numTermsUsed = nextTerm-1;

    % Cuatro residuos pedidos.
    RminusOnlyRaw = Y-Pminus;
    RplusOnlyRaw = Y-Pplus;
    RbothRaw = Y-Pminus-Pplus;

    % CORRECCION: la parte suave esta definida salvo constante aditiva.
    % Se ajustan [constante + portadora] conjuntamente en cada residuo.
    [constantMinus,cMinusResidual] = ...
        fitConstantAndCarrier(RminusOnlyRaw,carrier);

    [constantPlus,cPlusResidual] = ...
        fitConstantAndCarrier(RplusOnlyRaw,carrier);

    [jointConstant,cBothResidual] = ...
        fitConstantAndCarrier(RbothRaw,carrier);

    RminusOnly = RminusOnlyRaw-constantMinus;
    RplusOnly = RplusOnlyRaw-constantPlus;
    Rboth = RbothRaw-jointConstant;

    E = Rboth-cBothResidual*carrier;
    F = conj(carrier).*Rboth;
    Fc = F-cBothResidual;

    [constantPminus,projectionMinus] = ...
        fitConstantAndCarrier(Pminus,carrier);

    [constantPplus,projectionPlus] = ...
        fitConstantAndCarrier(Pplus,carrier);

    carrierCancellationIdentityError = abs( ...
        cBothResidual-(cY-projectionMinus-projectionPlus));

    constantCancellationIdentityError = abs( ...
        jointConstant-(constantY-constantPminus-constantPplus));

    maxConstantCancellationIdentityError = max( ...
        maxConstantCancellationIdentityError, ...
```

```
        constantCancellationIdentityError);

    % TEST 1: energias y termino cruzado.
    energyMinus = mean(abs(Pminus).^2);
    energyPlus = mean(abs(Pplus).^2);
    crossMinusPlus = 2*real(mean(conj(Pminus).*Pplus));
    energyCombined = mean(abs(Pminus+Pplus).^2);

    crossEnergyIdentityError = abs( ...
        energyCombined-(energyMinus+energyPlus+crossMinusPlus));

    % Forma centrada.
    PminusCentered = Pminus-mean(Pminus);
    PplusCentered = Pplus-mean(Pplus);

    centeredEnergyMinus = mean(abs(PminusCentered).^2);
    centeredEnergyPlus = mean(abs(PplusCentered).^2);

    centeredCross = 2*real(mean( ...
        conj(PminusCentered).*PplusCentered));

    centeredEnergyCombined = mean(abs( ...
        PminusCentered+PplusCentered).^2);

    % TEST 5: ecuacion funcional discreta y diferencial.
    dgamma = diff(gamma);

    derivativeDefect = diff(Rboth)./dgamma ...
        -1i*omega*0.5*(Rboth(2:end)+Rboth(1:end-1));

    discreteDefect = Rboth(2:end) ...
        -exp(1i*omega*dgamma).*Rboth(1:end-1);

    Fvariance = mean(abs(Fc).^2);
    FmaxDeviation = max(abs(Fc));
    FmeanAbsIncrement = mean(abs(diff(F)));
    FtotalVariation = sum(abs(diff(F)));
    Fanisotropy = complexAnisotropy(Fc,minimumVariance);

    corrGapReal = safeCorrelation(real(Fc),gapBefore);
    corrGapImag = safeCorrelation(imag(Fc),gapBefore);
    corrGapAbs = safeCorrelation(abs(Fc),gapBefore);

    corrGapResidualReal = safeCorrelation(real(Fc),gapResidual);
    corrGapResidualImag = safeCorrelation(imag(Fc),gapResidual);
    corrGapResidualAbs = safeCorrelation(abs(Fc),gapResidual);

    % Bloques de altura.
    heightEdges = round(linspace(1,N+1,numHeightBlocks+1));
    blockMeans = complex(NaN(numHeightBlocks,1));

    for hb = 1:numHeightBlocks

        idx = heightEdges(hb):(heightEdges(hb+1)-1);

        Fb = F(idx);
        Rb = Rboth(idx);
        blockMean = mean(Fb);
        blockMeans(hb) = blockMean;
        Fbc = Fb-blockMean;

        if numel(idx)>=2
            dgb = diff(gamma(idx));

            discreteBlock = Rb(2:end) ...
```

```
                -exp(1i*omega*dgb).*Rb(1:end-1);

            derivativeBlock = diff(Rb)./dgb ...
                -1i*omega*0.5*(Rb(2:end)+Rb(1:end-1));

            meanAbsDeltaBlock = mean(abs(diff(Fb)));

            variationPerHeight = sum(abs(diff(Fb))) ...
                /max(gamma(idx(end))-gamma(idx(1)),eps);

            discreteBlockRMS = sqrt(mean(abs(discreteBlock).^2));
            derivativeBlockRMS = sqrt(mean(abs(derivativeBlock).^2));
        else
            meanAbsDeltaBlock = NaN;
            variationPerHeight = NaN;
            discreteBlockRMS = NaN;
            derivativeBlockRMS = NaN;
        end

        blockRow = blockRow+1;

        blockRows(blockRow,:) = { ...
            Q,hb,indices(idx(1)),indices(idx(end)), ...
            gamma(idx(1)),gamma(idx(end)), ...
            real(blockMean),imag(blockMean),mean(abs(Fbc).^2), ...
            max(abs(Fb-cBothResidual)),max(abs(Fbc)), ...
            meanAbsDeltaBlock,variationPerHeight, ...
            complexAnisotropy(Fbc,minimumVariance), ...
            discreteBlockRMS,derivativeBlockRMS};
    end

    FblockMeanStd = sqrt(mean(abs(blockMeans-mean(blockMeans)).^2));

    % TEST 4: positividad elemental/Gram.
    normE2 = mean(abs(E).^2);
    normCarrier2 = mean(abs(carrier).^2);

    innerCarrierE = mean(conj(carrier).*E);
    innerEMinus = mean(conj(E).*Pminus);
    innerEPlus = mean(conj(E).*Pplus);
    innerMinusPlus = mean(conj(Pminus).*Pplus);

    carrierSchur = mean(abs(Rboth).^2) ...
        -abs(mean(conj(carrier).*Rboth))^2/normCarrier2;

    cauchyDetEMinus = normE2*energyMinus-abs(innerEMinus)^2;
    cauchyDetEPlus = normE2*energyPlus-abs(innerEPlus)^2;
    cauchyDetMinusPlus = energyMinus*energyPlus ...
        -abs(innerMinusPlus)^2;

    QEplusMinus = mean(abs(E+Pminus).^2);
    QEminusMinus = mean(abs(E-Pminus).^2);

    polarizationErrorMinus = abs( ...
        0.5*(QEplusMinus+QEminusMinus)-normE2-energyMinus);

    QEplusPlus = mean(abs(E+Pplus).^2);
    QEminusPlus = mean(abs(E-Pplus).^2);

    polarizationErrorPlus = abs( ...
        0.5*(QEplusPlus+QEminusPlus)-normE2-energyPlus);

    positivityRows(qi,:) = { ...
        Q,normE2,carrierSchur, ...
        abs(innerCarrierE),real(innerCarrierE),imag(innerCarrierE), ...
```

```
        real(innerEMinus),real(innerEPlus), ...
        cauchyDetEMinus,cauchyDetEPlus,cauchyDetMinusPlus, ...
        QEplusMinus,QEminusMinus,polarizationErrorMinus, ...
        QEplusPlus,QEminusPlus,polarizationErrorPlus};

    if numTermsUsed>0
        currentA = AQ(numTermsUsed);
        currentB = BQ(numTermsUsed);
    else
        currentA = 0;
        currentB = 0;
    end

    summaryRows(qi,:) = { ...
        Q,numTermsUsed, ...
        real(cMinusResidual),imag(cMinusResidual), ...
        real(cPlusResidual),imag(cPlusResidual), ...
        real(cBothResidual),imag(cBothResidual),abs(cBothResidual), ...
        abs(cBothResidual-0.5), ...
        real(jointConstant),imag(jointConstant), ...
        real(cBothResidual),imag(cBothResidual),abs(cBothResidual-0.5), ...
        real(projectionMinus),imag(projectionMinus), ...
        real(projectionPlus),imag(projectionPlus), ...
        carrierCancellationIdentityError, ...
        sqrt(mean(abs(RminusOnly).^2)), ...
        sqrt(mean(abs(RplusOnly).^2)), ...
        sqrt(mean(abs(Rboth).^2)),sqrt(normE2), ...
        energyMinus,energyPlus,crossMinusPlus,energyCombined, ...
        crossEnergyIdentityError, ...
        centeredEnergyMinus,centeredEnergyPlus,centeredCross, ...
        centeredEnergyCombined, ...
        Fvariance,FmaxDeviation,FmeanAbsIncrement,FtotalVariation, ...
        FblockMeanStd,Fanisotropy, ...
        sqrt(mean(abs(discreteDefect).^2)), ...
        sqrt(mean(abs(derivativeDefect).^2)), ...
        real(currentA),imag(currentA)};

    fprintf(['Q=%-6g terms=%-5d  c^-res=%.7f%+.7fi  ' ...
        'c^+res=%.7f%+.7fi  c^both=%.7f%+.7fi  ' ...
        'Var(F)=%.4e  RMS(E)=%.4e\n'], ...
        Q,numTermsUsed, ...
        real(cMinusResidual),imag(cMinusResidual), ...
        real(cPlusResidual),imag(cPlusResidual), ...
        real(cBothResidual),imag(cBothResidual), ...
        Fvariance,sqrt(normE2));

    if Q==Qmax
        RbothAtQmax = Rboth;
        FAtQmax = F;
        EAtQmax = E;
        PminusAtQmax = Pminus;
        PplusAtQmax = Pplus;
        cBothAtQmax = cBothResidual;
    end
end

summaryVariableNames = { ...
    'Q','num_prime_power_terms', ...
    'c_after_minus_real','c_after_minus_imag', ...
    'c_after_plus_real','c_after_plus_imag', ...
    'c_after_both_real','c_after_both_imag','c_after_both_abs', ...
    'distance_c_after_both_to_half', ...
    'joint_constant_real','joint_constant_imag', ...
    'joint_carrier_real','joint_carrier_imag', ...
    'distance_joint_carrier_to_half', ...
```

```
    'projection_Pminus_real','projection_Pminus_imag', ...
    'projection_Pplus_real','projection_Pplus_imag', ...
    'carrier_cancellation_identity_error', ...
    'rms_centered_Y_minus_Pminus','rms_centered_Y_minus_Pplus', ...
    'rms_centered_Y_minus_both', ...
    'rms_after_both_fitted_carrier', ...
    'energy_Pminus','energy_Pplus', ...
    'cross_term_2Re_inner_Pminus_Pplus', ...
    'energy_Pminus_plus_Pplus','cross_energy_identity_error', ...
    'centered_energy_Pminus','centered_energy_Pplus', ...
    'centered_cross_term','centered_energy_combined', ...
    'F_variance','F_max_abs_deviation_from_mean', ...
    'F_mean_abs_increment','F_total_variation', ...
    'F_block_mean_std','F_anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms', ...
    'A_Q_real','A_Q_imag'};

summaryTable = cell2table(summaryRows, ...
    'VariableNames',summaryVariableNames);

positivityVariableNames = { ...
    'Q','norm_E2','carrier_schur_complement', ...
    'carrier_projection_abs','carrier_projection_real', ...
    'carrier_projection_imag','Re_inner_E_Pminus', ...
    'Re_inner_E_Pplus','cauchy_det_E_Pminus', ...
    'cauchy_det_E_Pplus','cauchy_det_Pminus_Pplus', ...
    'Q_E_plus_Pminus','Q_E_minus_Pminus', ...
    'polarization_error_minus','Q_E_plus_Pplus', ...
    'Q_E_minus_Pplus','polarization_error_plus'};

positivityTable = cell2table(positivityRows, ...
    'VariableNames',positivityVariableNames);

blockVariableNames = { ...
    'Q','block','k_start','k_end','gamma_start','gamma_end', ...
    'F_mean_real','F_mean_imag','F_variance', ...
    'sup_abs_F_minus_global_c','sup_abs_F_minus_block_c', ...
    'mean_abs_delta_F','variation_per_height','anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms'};

blockTable = cell2table(blockRows(1:blockRow,:), ...
    'VariableNames',blockVariableNames);

%% ========================================================================
% DISPLAY PRINCIPAL: TEST 1 Y TEST 5
% ========================================================================

fprintf('\n============================================================\n');
fprintf('TEST 1: RAMAS; TEST 5: ECUACION FUNCIONAL\n');
fprintf('============================================================\n');

mainDisplayVariables = { ...
    'Q','num_prime_power_terms', ...
    'c_after_minus_real','c_after_minus_imag', ...
    'c_after_plus_real','c_after_plus_imag', ...
    'c_after_both_real','c_after_both_imag', ...
    'distance_c_after_both_to_half', ...
    'rms_centered_Y_minus_Pminus','rms_centered_Y_minus_Pplus', ...
    'rms_centered_Y_minus_both', ...
    'rms_after_both_fitted_carrier', ...
    'energy_Pminus','energy_Pplus', ...
    'cross_term_2Re_inner_Pminus_Pplus', ...
    'F_variance','F_max_abs_deviation_from_mean', ...
    'F_block_mean_std','F_anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms'};
```

```
704
705 disp(summaryTable(:,mainDisplayVariables));
706
707 fprintf('max error R_j-R_{j-1}=-P_j                  : %.6e\n', ...
708     maxIncrementIdentityError);
709
710 fprintf('max error ley de energia de ramas           : %.6e\n', ...
711     max(summaryTable.cross_energy_identity_error));
712
713 fprintf('max error identidad de proyeccion portadora : %.6e\n', ...
714     max(summaryTable.carrier_cancellation_identity_error));
715 fprintf('max error identidad de constantes            : %.6e\n', ...
716     maxConstantCancellationIdentityError);
717
718 %% ========================================================================
719 % TEST 3: TELESCOPADO DENTRO DE CADA PRIMO p
720 % ========================================================================
721
722 [groupSummaryTable,groupPartialTable] = primePowerTelescopingTests( ...
723     qVec,basePrimeVec,powerExponentVec,cMinusVec,cPlusVec, ...
724     gamma,carrier);
725
726 fprintf('\n============================================================\n');
727 fprintf('TEST 3: TELESCOPADO EN p,p^2,...,p^R\n');
728 fprintf('============================================================\n');
729
730 if isempty(groupSummaryTable)
731     fprintf('No hay primos con al menos dos potencias <=Qmax.\n');
732 else
733     [~,orderTelescoping] = sort( ...
734         groupSummaryTable.boundary_fit_relative_rms_combined,'ascend');
735
736     numDisplay = min(20,height(groupSummaryTable));
737
738     telescopingDisplayVariables = { ...
739         'base_prime','num_powers','largest_power', ...
740         'combined_energy_ratio_sum_over_individual', ...
741         'combined_rms_over_sum_term_rms', ...
742         'boundary_fit_relative_rms_combined', ...
743         'boundary_fit_relative_rms_minus', ...
744         'boundary_fit_relative_rms_plus', ...
745         'boundary_alpha_first_real','boundary_alpha_first_imag', ...
746         'boundary_alpha_last_real','boundary_alpha_last_imag', ...
747         'tail_primitive_start_abs_minus', ...
748         'tail_primitive_start_abs_plus'};
749
750     disp(groupSummaryTable( ...
751         orderTelescoping(1:numDisplay), ...
752         telescopingDisplayVariables));
753
754     fprintf(['boundary_fit_relative_rms_combined cercano a cero seria ' ...
755         'evidencia de colapso a los dos bordes del bloque p^r.\n']);
756 end
757
758 %% ========================================================================
759 % TEST 4: FUNCIONALES POSITIVOS Y MATRICES DE GRAM
760 % ========================================================================
761
762 Gminus = double(gramMinus);
763 Gplus = double(gramPlus);
764
765 GcombinedCentered = Gminus+Gplus;
766 GcombinedCentered = GcombinedCentered-mean(GcombinedCentered,1);
767
768 GbranchCentered = [ ...
```

```
    Gminus-mean(Gminus,1), ...
    Gplus-mean(Gplus,1)];

gramCombined = (GcombinedCentered'*GcombinedCentered)/numGramPoints;
gramBranch = (GbranchCentered'*GbranchCentered)/numGramPoints;

gramCombined = 0.5*(gramCombined+gramCombined');
gramBranch = 0.5*(gramBranch+gramBranch');

eigCombined = sort(real(eig(gramCombined)));
eigBranch = sort(real(eig(gramBranch)));

fprintf('\n============================================================\n');
fprintf('TEST 4: POSITIVIDAD Y GRAM\n');
fprintf('============================================================\n');

positivityDisplayVariables = { ...
    'Q','norm_E2','carrier_schur_complement', ...
    'carrier_projection_abs','Re_inner_E_Pminus', ...
    'Re_inner_E_Pplus','cauchy_det_E_Pminus', ...
    'cauchy_det_E_Pplus','cauchy_det_Pminus_Pplus', ...
    'polarization_error_minus','polarization_error_plus'};

disp(positivityTable(:,positivityDisplayVariables));

fprintf('Gram combinada: min=%.6e, max=%.6e, negativos<-1e-10=%d\n', ...
    min(eigCombined),max(eigCombined),nnz(eigCombined<-1e-10));

fprintf('Gram por ramas: min=%.6e, max=%.6e, negativos<-1e-10=%d\n', ...
    min(eigBranch),max(eigBranch),nnz(eigBranch<-1e-10));

%% ========================================================================
% TEST 5 POR BLOQUES PARA Qmax
% ========================================================================

blockQmax = blockTable(blockTable.Q==Qmax,:);

fprintf('\n============================================================\n');
fprintf('TEST 5: F_Q POR BLOQUES DE ALTURA, Q=%g\n',Qmax);
fprintf('============================================================\n');

blockDisplayVariables = { ...
    'block','gamma_start','gamma_end', ...
    'F_mean_real','F_mean_imag','F_variance', ...
    'sup_abs_F_minus_global_c','sup_abs_F_minus_block_c', ...
    'mean_abs_delta_F','variation_per_height','anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms'};

disp(blockQmax(:,blockDisplayVariables));

fprintf('Rango Re media por bloque : [%.12g, %.12g]\n', ...
    min(blockQmax.F_mean_real),max(blockQmax.F_mean_real));

fprintf('Rango Im media por bloque : [%.12g, %.12g]\n', ...
    min(blockQmax.F_mean_imag),max(blockQmax.F_mean_imag));

fprintf('Max sup global por bloque : %.6e\n', ...
    max(blockQmax.sup_abs_F_minus_global_c));

fprintf('Max sup local por bloque  : %.6e\n', ...
    max(blockQmax.sup_abs_F_minus_block_c));

%% ========================================================================
% TEST 6: FRECUENCIAS OMITIDAS
% ========================================================================
```

```
834
835 spectralIndices = unique(round(linspace(1,N, ...
836     min(N,maxSpectralPoints)))).';
837
838 Ts = gamma(spectralIndices);
839 Fs = FAtQmax(spectralIndices)-cBothAtQmax;
840 Es = EAtQmax(spectralIndices);
841
842 testStart = Qmax+1;
843 testEnd = Qmax+omittedTestWidth;
844 nCandidates = (testStart:testEnd).';
845
846 [qTestAll,pTestAll,rTestAll,lambdaTestAll] = makePrimePowers(testEnd);
847
848 [isPrimePowerCandidate,locPP] = ismember(nCandidates,qTestAll);
849
850 candidateLambda = zeros(size(nCandidates));
851 candidateBasePrime = zeros(size(nCandidates));
852 candidateExponent = zeros(size(nCandidates));
853
854 candidateLambda(isPrimePowerCandidate) = ...
855     lambdaTestAll(locPP(isPrimePowerCandidate));
856
857 candidateBasePrime(isPrimePowerCandidate) = ...
858     pTestAll(locPP(isPrimePowerCandidate));
859
860 candidateExponent(isPrimePowerCandidate) = ...
861     rTestAll(locPP(isPrimePowerCandidate));
862
863 omittedRows = cell(numel(nCandidates),19);
864
865 for jj = 1:numel(nCandidates)
866
867     n = nCandidates(jj);
868     ell = log(n);
869
870     basisMinusF = exp(-1i*ell*Ts);
871     basisPlusF = exp(1i*ell*Ts);
872
873     fitMinusF = fitCarrier(Fs,basisMinusF);
874     fitPlusF = fitCarrier(Fs,basisPlusF);
875
876     basisMinusR = exp(1i*(omega-ell)*Ts);
877     basisPlusR = exp(1i*(omega+ell)*Ts);
878
879     fitMinusR = fitCarrier(Es,basisMinusR);
880     fitPlusR = fitCarrier(Es,basisPlusR);
881
882     theoryMinus = 0;
883     theoryPlus = 0;
884
885     if isPrimePowerCandidate(jj)
886         lam = candidateLambda(jj);
887
888         theoryMinus = 1i*lam / ...
889             (2*pi*sqrt(n)*(omega-ell));
890
891         theoryPlus = 1i*lam / ...
892             (2*pi*sqrt(n)*(omega+ell));
893     end
894
895     omittedRows(jj,:) = { ...
896         n,isPrimePowerCandidate(jj),candidateBasePrime(jj), ...
897         candidateExponent(jj),candidateLambda(jj), ...
898         real(fitMinusF),imag(fitMinusF),abs(fitMinusF), ...
```

```
        real(fitPlusF),imag(fitPlusF),abs(fitPlusF), ...
        real(fitMinusR),imag(fitMinusR), ...
        real(fitPlusR),imag(fitPlusR), ...
        real(theoryMinus),imag(theoryMinus), ...
        real(theoryPlus),imag(theoryPlus)};
end

omittedVariableNames = { ...
    'n','is_prime_power','base_prime','power_exponent','Lambda_n', ...
    'fit_F_minus_real','fit_F_minus_imag','fit_F_minus_abs', ...
    'fit_F_plus_real','fit_F_plus_imag','fit_F_plus_abs', ...
    'fit_R_minus_real','fit_R_minus_imag', ...
    'fit_R_plus_real','fit_R_plus_imag', ...
    'theory_minus_real','theory_minus_imag', ...
    'theory_plus_real','theory_plus_imag'};

omittedTable = cell2table(omittedRows, ...
    'VariableNames',omittedVariableNames);

ppMask = omittedTable.is_prime_power;
nullMask = ~ppMask;

medianPPMinus = median(omittedTable.fit_F_minus_abs(ppMask));
medianNullMinus = median(omittedTable.fit_F_minus_abs(nullMask));

medianPPPlus = median(omittedTable.fit_F_plus_abs(ppMask));
medianNullPlus = median(omittedTable.fit_F_plus_abs(nullMask));

omittedPPMask = qTestAll>Qmax & qTestAll<=testEnd;
omittedPP = qTestAll(omittedPPMask);
omittedPPLambda = lambdaTestAll(omittedPPMask);

Xomitted = complex(zeros(numel(Ts),2*numel(omittedPP)));

for jj = 1:numel(omittedPP)
    ell = log(omittedPP(jj));
    Xomitted(:,jj) = exp(-1i*ell*Ts);
    Xomitted(:,numel(omittedPP)+jj) = exp(1i*ell*Ts);
end

if isempty(Xomitted)
    fittedOmittedCoefficients = [];
    FsAfterOmittedFit = Fs;
    omittedExplainedFraction = NaN;
    fittedTheoryCorrelation = NaN;
    relativeTheoryCoefficientError = NaN;
else
    fittedOmittedCoefficients = Xomitted\Fs;
    FsAfterOmittedFit = Fs-Xomitted*fittedOmittedCoefficients;

    omittedExplainedFraction = 1 ...
        -sum(abs(FsAfterOmittedFit).^2)/sum(abs(Fs).^2);

    theoryOmittedCoefficients = complex(zeros(2*numel(omittedPP),1));

    for jj = 1:numel(omittedPP)

        n = omittedPP(jj);
        ell = log(n);
        lam = omittedPPLambda(jj);

        theoryOmittedCoefficients(jj) = 1i*lam / ...
            (2*pi*sqrt(n)*(omega-ell));

        theoryOmittedCoefficients(numel(omittedPP)+jj) = 1i*lam / ...
```

```
            (2*pi*sqrt(n)*(omega+ell));
    end

    fittedTheoryCorrelation = complexVectorCorrelation( ...
        fittedOmittedCoefficients,theoryOmittedCoefficients);

    relativeTheoryCoefficientError = norm( ...
        fittedOmittedCoefficients-theoryOmittedCoefficients) ...
        /max(norm(theoryOmittedCoefficients),eps);
end

nuMax = log(testEnd);
nuGrid = linspace(-nuMax,nuMax,numFrequencyGridPoints).';
frequencyAmplitude = zeros(size(nuGrid));

for jj = 1:numel(nuGrid)
    basis = exp(1i*nuGrid(jj)*Ts);
    frequencyAmplitude(jj) = abs(fitCarrier(Fs,basis));
end

[sortedAmplitude,sortOrder] = sort(frequencyAmplitude,'descend');
topCount = min(30,numel(sortOrder));

topFrequencyTable = table( ...
    (1:topCount).', ...
    nuGrid(sortOrder(1:topCount)), ...
    sortedAmplitude(1:topCount), ...
    'VariableNames',{'rank','demodulated_frequency','fit_amplitude'});

uniquenessTable = table( ...
    Qmax,numel(omittedPP), ...
    medianPPMinus,medianNullMinus,medianPPMinus/medianNullMinus, ...
    medianPPPlus,medianNullPlus,medianPPPlus/medianNullPlus, ...
    omittedExplainedFraction, ...
    real(fittedTheoryCorrelation),imag(fittedTheoryCorrelation), ...
    abs(fittedTheoryCorrelation),relativeTheoryCoefficientError, ...
    mean(abs(Fs).^2),mean(abs(FsAfterOmittedFit).^2), ...
    complexAnisotropy(Fs,minimumVariance), ...
    safeCorrelation(real(Fs),gapBefore(spectralIndices)), ...
    safeCorrelation(imag(Fs),gapBefore(spectralIndices)), ...
    safeCorrelation(abs(Fs),gapBefore(spectralIndices)), ...
    'VariableNames',{ ...
    'Qmax','num_omitted_prime_powers_joint_fit', ...
    'median_minus_amplitude_prime_power', ...
    'median_minus_amplitude_null', ...
    'minus_prime_power_to_null_ratio', ...
    'median_plus_amplitude_prime_power', ...
    'median_plus_amplitude_null', ...
    'plus_prime_power_to_null_ratio', ...
    'joint_omitted_explained_fraction', ...
    'fit_theory_correlation_real','fit_theory_correlation_imag', ...
    'fit_theory_correlation_abs','relative_theory_coefficient_error', ...
    'F_energy_before_omitted_fit','F_energy_after_omitted_fit', ...
    'F_anisotropy','corr_gap_realF','corr_gap_imagF','corr_gap_absF'});

fprintf('\n============================================================\n');
fprintf('TEST 6: UNICIDAD Y FRECUENCIAS OMITIDAS\n');
fprintf('============================================================\n');

disp(uniquenessTable);

omittedPrimePowerOnly = omittedTable(omittedTable.is_prime_power,:);

if ~isempty(omittedPrimePowerOnly)

```

```
    omittedPrimePowerOnly.max_branch_fit_abs = max( ...
        omittedPrimePowerOnly.fit_F_minus_abs, ...
        omittedPrimePowerOnly.fit_F_plus_abs);

    omittedPrimePowerOnly = sortrows( ...
        omittedPrimePowerOnly,'max_branch_fit_abs','descend');

    numDisplay = min(20,height(omittedPrimePowerOnly));

    fprintf('\nPotencias primas omitidas con mayor amplitud:\n');

    disp(omittedPrimePowerOnly(1:numDisplay,{ ...
        'n','base_prime','power_exponent','Lambda_n', ...
        'fit_F_minus_real','fit_F_minus_imag','fit_F_minus_abs', ...
        'fit_F_plus_real','fit_F_plus_imag','fit_F_plus_abs', ...
        'theory_minus_real','theory_minus_imag', ...
        'theory_plus_real','theory_plus_imag', ...
        'max_branch_fit_abs'}));
end

fprintf('\nFrecuencias no nulas dominantes de F_Q-mean(F_Q):\n');
disp(topFrequencyTable);

fprintf('\nDIAGNOSTICO FINAL PARA Q=%g\n',Qmax);
fprintf('mean(F_Q)                          = %.12g %+.12gi\n', ...
    real(cBothAtQmax),imag(cBothAtQmax));
fprintf('|mean(F_Q)-1/2|                    = %.6e\n', ...
    abs(cBothAtQmax-0.5));
fprintf('Var(F_Q)                           = %.6e\n', ...
    mean(abs(FAtQmax-cBothAtQmax).^2));
fprintf('sup |F_Q-mean(F_Q)|                = %.6e\n', ...
    max(abs(FAtQmax-cBothAtQmax)));
fprintf('RMS E_Q                            = %.6e\n', ...
    sqrt(mean(abs(EAtQmax).^2)));
fprintf('razon omitidas potencia/null (-) = %.6f\n', ...
    medianPPMinus/medianNullMinus);
fprintf('razon omitidas potencia/null (+) = %.6f\n', ...
    medianPPPlus/medianNullPlus);
fprintf('fraccion explicada ajuste omitido= %.6f\n', ...
    omittedExplainedFraction);

%% ========================================================================
% FIGURAS: NO SE CIERRA NINGUNA
% ========================================================================

figure('Name','Coeficientes de portadora por rama','Color','w');
tiledlayout(2,1,'TileSpacing','compact','Padding','compact');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_minus_real,'-o','LineWidth',1.2);
hold on;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_plus_real,'-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_both_real,'-o','LineWidth',1.2);
yline(0.5,':');
grid on;
xlabel('Q');
ylabel('Parte real');
legend('Y-P^-','Y-P^+','Y-P^--P^+','1/2','Location','best');
title('Coeficiente ajustado de exp(i omega T)');

nexttile;
semilogx(max(summaryTable.Q,1), ...
```

```
    summaryTable.c_after_minus_imag,'-o','LineWidth',1.2);
hold on;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_plus_imag,'-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_both_imag,'-o','LineWidth',1.2);
yline(0,':');
grid on;
xlabel('Q');
ylabel('Parte imaginaria');
legend('Y-P^-','Y-P^+','Y-P^--P^+','0','Location','best');

figure('Name','Energia y termino cruzado de ramas','Color','w');
semilogx(max(summaryTable.Q,1), ...
    summaryTable.energy_Pminus,'-o','LineWidth',1.2);
hold on;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.energy_Pplus,'-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.cross_term_2Re_inner_Pminus_Pplus, ...
    '-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.energy_Pminus_plus_Pplus,'-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('Energia');
legend('||P^-||^2','||P^+||^2','2Re<P^-,P^+>', ...
    '||P^-+P^+||^2','Location','best');
title('Cancelacion entre ramas');

figure('Name','Ecuacion funcional del residuo','Color','w');
tiledlayout(2,2,'TileSpacing','compact','Padding','compact');

nexttile;
semilogx(max(summaryTable.Q,1),summaryTable.F_variance, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('Var(F_Q)');
title('Dependencia residual de T');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.distance_c_after_both_to_half, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('|mean(F_Q)-1/2|');
title('Distancia del coeficiente a 1/2');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.carrier_discrete_defect_rms, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('RMS defecto discreto');
title('R_{k+1}-e^{i omega Delta T}R_k');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.F_block_mean_std, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
```

```
ylabel('STD de medias por bloque');
title('Deriva por altura');

figure('Name','Sumas parciales A_Q y B_Q','Color','w');
tiledlayout(2,1,'TileSpacing','compact','Padding','compact');

nexttile;
plot(qVec,real(AQ),'LineWidth',1);
hold on;
plot(qVec,imag(AQ),'LineWidth',1);
yline(0.5,':');
yline(-0.5,':');
set(gca,'XScale','log');
grid on;
xlabel('Q');
ylabel('A_Q');
legend('Re A_Q','Im A_Q','1/2','-1/2','Location','best');

nexttile;
plot(qVec,real(BQ),'LineWidth',1);
hold on;
plot(qVec,imag(BQ),'LineWidth',1);
yline(0.5,':');
yline(-0.5,':');
set(gca,'XScale','log');
grid on;
xlabel('Q');
ylabel('B_Q');
legend('Re B_Q','Im B_Q','1/2','-1/2','Location','best');

blockSupMatrix = reshape( ...
    blockTable.sup_abs_F_minus_global_c, ...
    numHeightBlocks,numel(Qlist)).';

figure('Name','Desviacion suprema por bloques','Color','w');
imagesc(1:numHeightBlocks,1:numel(Qlist),log10(blockSupMatrix));
colorbar;
yticks(1:numel(Qlist));
yticklabels(string(Qlist));
xlabel('Bloque de altura');
ylabel('Q');
title('log10 sup_{T en I_j}|F_Q(T)-c_Q|');

figure('Name','Eigenvalores de Gram','Color','w');
tiledlayout(1,2,'TileSpacing','compact','Padding','compact');

nexttile;
semilogy(max(eigCombined,eps),'-o');
grid on;
xlabel('Indice');
ylabel('Eigenvalor');
title('Gram combinada');

nexttile;
semilogy(max(eigBranch,eps),'-o');
grid on;
xlabel('Indice');
ylabel('Eigenvalor');
title('Gram por ramas');

figure('Name','Frecuencias omitidas','Color','w');
semilogy(omittedTable.n,omittedTable.fit_F_minus_abs,'.');
hold on;
semilogy(omittedTable.n(ppMask), ...
    omittedTable.fit_F_minus_abs(ppMask),'o');
```

```
semilogy(omittedTable.n,omittedTable.fit_F_plus_abs,'.');
semilogy(omittedTable.n(ppMask), ...
    omittedTable.fit_F_plus_abs(ppMask),'s');
grid on;
xlabel('n');
ylabel('Amplitud ajustada');
legend('Rama - todos','Rama - potencias primas', ...
    'Rama + todos','Rama + potencias primas','Location','best');

figure('Name','Espectro no nulo de F_Q','Color','w');
plot(nuGrid,frequencyAmplitude,'LineWidth',1);
grid on;
xlabel('nu en exp(i nu T)');
ylabel('Amplitud');
title(sprintf('F_Q-mean(F_Q), Q=%g',Qmax));

%% ========================================================================
% GUARDADO OPCIONAL; NUNCA ES NECESARIO PARA EJECUTAR
% ========================================================================

if saveResults

    if ~isfolder(outputDirectory)
        mkdir(outputDirectory);
    end

    writetable(summaryTable, ...
        fullfile(outputDirectory,'SIX_TESTS_SUMMARY.csv'));

    writetable(coefficientCutoffDisplay, ...
        fullfile(outputDirectory,'COEFFICIENT_CUTOFF_SUMMARY.csv'));

    writetable(groupSummaryTable, ...
        fullfile(outputDirectory,'TELESCOPING_SUMMARY.csv'));

    writetable(groupPartialTable, ...
        fullfile(outputDirectory,'TELESCOPING_PARTIALS.csv'));

    writetable(positivityTable, ...
        fullfile(outputDirectory,'POSITIVITY_FUNCTIONALS.csv'));

    writetable(blockTable, ...
        fullfile(outputDirectory,'BLOCKWISE_FUNCTIONAL_EQUATION.csv'));

    writetable(omittedTable, ...
        fullfile(outputDirectory,'OMITTED_FREQUENCIES.csv'));

    writetable(uniquenessTable, ...
        fullfile(outputDirectory,'UNIQUENESS_SUMMARY.csv'));

    save(fullfile(outputDirectory,'ALL_RESULTS.mat'), ...
        'summaryTable','coefficientCutoffDisplay','radialAuditTable', ...
        'groupSummaryTable','groupPartialTable', ...
        'positivityTable','blockTable','omittedTable', ...
        'uniquenessTable','topFrequencyTable', ...
        'gramCombined','gramBranch','eigCombined','eigBranch', ...
        'RbothAtQmax','FAtQmax','EAtQmax', ...
        'PminusAtQmax','PplusAtQmax','cBothAtQmax','-v7.3');

    fprintf('\nResultados guardados en %s\n',outputDirectory);
end

fprintf('\n============================================================\n');
fprintf('TERMINADO\n');
fprintf('Tablas disponibles directamente en el workspace:\n');
```

```
fprintf('  radialAuditTable\n');
fprintf('  summaryTable\n');
fprintf('  coefficientCutoffDisplay\n');
fprintf('  groupSummaryTable\n');
fprintf('  groupPartialTable\n');
fprintf('  positivityTable\n');
fprintf('  blockQmax\n');
fprintf('  omittedTable\n');
fprintf('  uniquenessTable\n');
fprintf('  topFrequencyTable\n');
fprintf('============================================================\n');

%% ========================================================================
% FUNCIONES LOCALES
% ========================================================================

function c = fitCarrier(z,b)

    z = z(:);
    b = b(:);

    denominator = b'*b;

    if abs(denominator)<=eps
        c = NaN;
    else
        c = (b'*z)/denominator;
    end
end

function [a,c] = fitConstantAndCarrier(z,b)

    z = z(:);
    b = b(:);

    X = [ones(size(b)),b];
    coefficients = X\z;

    a = coefficients(1);
    c = coefficients(2);
end

function value = complexAnisotropy(z,minimumVariance)

    z = z(:);
    z = z-mean(z,'omitnan');

    denominator = mean(abs(z).^2,'omitnan');

    if ~isfinite(denominator) || denominator<=minimumVariance
        value = NaN;
    else
        value = abs(mean(z.^2,'omitnan'))/denominator;
    end
end

function r = safeCorrelation(x,y)

    x = x(:);
    y = y(:);

    valid = isfinite(x) & isfinite(y);

    if nnz(valid)<3
        r = NaN;
```

```
        return;
    end

    x = x(valid);
    y = y(valid);

    x = x-mean(x);
    y = y-mean(y);

    denominator = sqrt(sum(abs(x).^2)*sum(abs(y).^2));

    if denominator<=eps
        r = NaN;
    else
        r = real((x'*y)/denominator);
    end
end

function r = complexVectorCorrelation(x,y)

    x = x(:);
    y = y(:);

    valid = isfinite(real(x)) & isfinite(imag(x)) ...
        & isfinite(real(y)) & isfinite(imag(y));

    x = x(valid);
    y = y(valid);

    denominator = norm(x)*norm(y);

    if denominator<=eps
        r = NaN;
    else
        r = (x'*y)/denominator;
    end
end

function [qVec,basePrimeVec,powerExponentVec,lambdaVec] = ...
    makePrimePowers(Qmax)

    if Qmax<2
        qVec = zeros(0,1);
        basePrimeVec = zeros(0,1);
        powerExponentVec = zeros(0,1);
        lambdaVec = zeros(0,1);
        return;
    end

    pList = primes(Qmax);

    qList = zeros(0,1);
    baseList = zeros(0,1);
    exponentList = zeros(0,1);
    lambdaList = zeros(0,1);

    for jj = 1:numel(pList)

        p = pList(jj);
        currentPower = p;
        exponent = 1;

        while currentPower<=Qmax

            qList(end+1,1) = currentPower; %#ok<AGROW>
```

```
            baseList(end+1,1) = p; %#ok<AGROW>
            exponentList(end+1,1) = exponent; %#ok<AGROW>
            lambdaList(end+1,1) = log(p); %#ok<AGROW>

            if currentPower>floor(Qmax/p)
                break;
            end

            currentPower = currentPower*p;
            exponent = exponent+1;
        end
    end

    [qVec,order] = sort(qList);
    basePrimeVec = baseList(order);
    powerExponentVec = exponentList(order);
    lambdaVec = lambdaList(order);

    [qVec,uniqueIndex] = unique(qVec,'stable');
    basePrimeVec = basePrimeVec(uniqueIndex);
    powerExponentVec = powerExponentVec(uniqueIndex);
    lambdaVec = lambdaVec(uniqueIndex);
end

function [summaryTable,partialTable] = primePowerTelescopingTests( ...
    qVec,basePrimeVec,powerExponentVec,cMinusVec,cPlusVec, ...
    gamma,carrier)

    uniquePrimes = unique(basePrimeVec);

    keep = false(size(uniquePrimes));

    for jj = 1:numel(uniquePrimes)
        keep(jj) = nnz(basePrimeVec==uniquePrimes(jj))>=2;
    end

    uniquePrimes = uniquePrimes(keep);

    summaryNames = { ...
        'base_prime','num_powers','largest_power', ...
        'combined_energy_ratio_sum_over_individual', ...
        'minus_energy_ratio_sum_over_individual', ...
        'plus_energy_ratio_sum_over_individual', ...
        'combined_rms_over_sum_term_rms', ...
        'boundary_fit_relative_rms_combined', ...
        'boundary_fit_relative_rms_minus', ...
        'boundary_fit_relative_rms_plus', ...
        'boundary_alpha_first_real','boundary_alpha_first_imag', ...
        'boundary_alpha_last_real','boundary_alpha_last_imag', ...
        'tail_primitive_error_minus','tail_primitive_error_plus', ...
        'tail_primitive_start_abs_minus', ...
        'tail_primitive_start_abs_plus'};

    partialNames = { ...
        'base_prime','power_exponent','n_prime_power', ...
        'term_rms_minus','term_rms_plus','term_rms_combined', ...
        'partial_rms_minus','partial_rms_plus','partial_rms_combined', ...
        'partial_coefficient_minus_real', ...
        'partial_coefficient_minus_imag', ...
        'partial_coefficient_plus_real', ...
        'partial_coefficient_plus_imag'};

    if isempty(uniquePrimes)
        summaryTable = cell2table(cell(0,numel(summaryNames)), ...
            'VariableNames',summaryNames);
```

```
1484
1485        partialTable = cell2table(cell(0,numel(partialNames)), ...
1486            'VariableNames',partialNames);
1487
1488        return;
1489    end
1490
1491    totalPartialRows = 0;
1492
1493    for jj = 1:numel(uniquePrimes)
1494        totalPartialRows = totalPartialRows ...
1495            +nnz(basePrimeVec==uniquePrimes(jj));
1496    end
1497
1498    summaryRows = cell(numel(uniquePrimes),numel(summaryNames));
1499    partialRows = cell(totalPartialRows,numel(partialNames));
1500
1501    partialRow = 0;
1502
1503    for jp = 1:numel(uniquePrimes)
1504
1505        p = uniquePrimes(jp);
1506        idxTerms = find(basePrimeVec==p);
1507
1508        [~,order] = sort(powerExponentVec(idxTerms));
1509        idxTerms = idxTerms(order);
1510
1511        R = numel(idxTerms);
1512
1513        sumMinus = complex(zeros(size(gamma)));
1514        sumPlus = complex(zeros(size(gamma)));
1515
1516        sumEnergyMinus = 0;
1517        sumEnergyPlus = 0;
1518        sumEnergyCombined = 0;
1519        sumTermRMSCombined = 0;
1520
1521        firstMinus = [];
1522        firstPlus = [];
1523        lastMinus = [];
1524        lastPlus = [];
1525
1526        cumulativeCoefficientMinus = 0;
1527        cumulativeCoefficientPlus = 0;
1528
1529        for rr = 1:R
1530
1531            j = idxTerms(rr);
1532            ell = log(qVec(j));
1533            z = exp(1i*ell*gamma);
1534
1535            termMinus = cMinusVec(j).*carrier.*conj(z);
1536            termPlus = cPlusVec(j).*carrier.*z;
1537            termCombined = termMinus+termPlus;
1538
1539            if rr==1
1540                firstMinus = termMinus;
1541                firstPlus = termPlus;
1542            end
1543
1544            if rr==R
1545                lastMinus = termMinus;
1546                lastPlus = termPlus;
1547            end
1548
```

```
            sumMinus = sumMinus+termMinus;
            sumPlus = sumPlus+termPlus;

            termEnergyMinus = mean(abs(termMinus).^2);
            termEnergyPlus = mean(abs(termPlus).^2);
            termEnergyCombined = mean(abs(termCombined).^2);

            sumEnergyMinus = sumEnergyMinus+termEnergyMinus;
            sumEnergyPlus = sumEnergyPlus+termEnergyPlus;
            sumEnergyCombined = sumEnergyCombined+termEnergyCombined;

            sumTermRMSCombined = sumTermRMSCombined ...
                +sqrt(termEnergyCombined);

            cumulativeCoefficientMinus = cumulativeCoefficientMinus ...
                +cMinusVec(j);

            cumulativeCoefficientPlus = cumulativeCoefficientPlus ...
                +cPlusVec(j);

            partialRow = partialRow+1;

            partialRows(partialRow,:) = { ...
                p,powerExponentVec(j),qVec(j), ...
                sqrt(termEnergyMinus),sqrt(termEnergyPlus), ...
                sqrt(termEnergyCombined), ...
                sqrt(mean(abs(sumMinus).^2)), ...
                sqrt(mean(abs(sumPlus).^2)), ...
                sqrt(mean(abs(sumMinus+sumPlus).^2)), ...
                real(cumulativeCoefficientMinus), ...
                imag(cumulativeCoefficientMinus), ...
                real(cumulativeCoefficientPlus), ...
                imag(cumulativeCoefficientPlus)};
        end

        sumCombined = sumMinus+sumPlus;

        energyRatioCombined = mean(abs(sumCombined).^2) ...
            /max(sumEnergyCombined,eps);

        energyRatioMinus = mean(abs(sumMinus).^2) ...
            /max(sumEnergyMinus,eps);

        energyRatioPlus = mean(abs(sumPlus).^2) ...
            /max(sumEnergyPlus,eps);

        rmsOverSumRMS = sqrt(mean(abs(sumCombined).^2)) ...
            /max(sumTermRMSCombined,eps);

        firstCombined = firstMinus+firstPlus;
        lastCombined = lastMinus+lastPlus;

        [relativeCombined,alphaCombined] = boundaryFit( ...
            sumCombined,firstCombined,lastCombined);

        [relativeMinus,~] = boundaryFit( ...
            sumMinus,firstMinus,lastMinus);

        [relativePlus,~] = boundaryFit( ...
            sumPlus,firstPlus,lastPlus);

        cM = cMinusVec(idxTerms);
        cP = cPlusVec(idxTerms);

        % Toda sucesion admite esta primitiva de cola. La identidad es
```

```
        % tautologica; lo informativo es su tamano y si existe forma cerrada.
        aM = [-flipud(cumsum(flipud(cM)));0];
        aP = [-flipud(cumsum(flipud(cP)));0];

        primitiveErrorM = max(abs(cM-(aM(2:end)-aM(1:end-1))));
        primitiveErrorP = max(abs(cP-(aP(2:end)-aP(1:end-1))));

        summaryRows(jp,:) = { ...
            p,R,max(qVec(idxTerms)), ...
            energyRatioCombined,energyRatioMinus,energyRatioPlus, ...
            rmsOverSumRMS, ...
            relativeCombined,relativeMinus,relativePlus, ...
            real(alphaCombined(1)),imag(alphaCombined(1)), ...
            real(alphaCombined(2)),imag(alphaCombined(2)), ...
            primitiveErrorM,primitiveErrorP, ...
            abs(aM(1)),abs(aP(1))};
    end

    summaryTable = cell2table(summaryRows, ...
        'VariableNames',summaryNames);

    partialTable = cell2table(partialRows(1:partialRow,:), ...
        'VariableNames',partialNames);
end

function [relativeRMS,alpha] = boundaryFit(target,firstTerm,lastTerm)

    target = target(:);
    firstTerm = firstTerm(:);
    lastTerm = lastTerm(:);

    X = [firstTerm,lastTerm];
    alpha = X\target;

    residual = target-X*alpha;

    relativeRMS = sqrt(mean(abs(residual).^2)) ...
        /max(sqrt(mean(abs(target).^2)),eps);
end


function [radius,centers,densitySmooth,peakRadii] = ...
    radialDistributionSummary(z,numBins,smoothWidth)

    z = z(:);
    z = z(isfinite(real(z)) & isfinite(imag(z)));

    center = mean(z);
    radius = abs(z-center);

    rMin = min(radius);
    rMax = max(radius);

    if rMax<=rMin
        rMax = rMin+eps(rMin+1);
    end

    edges = linspace(rMin,rMax,numBins+1);
    density = histcounts(radius,edges,'Normalization','pdf');
    centers = ((edges(1:end-1)+edges(2:end))/2).';

    densitySmooth = smoothdata(density(:),'gaussian',smoothWidth);

    candidates = find( ...
        densitySmooth(2:end-1)>densitySmooth(1:end-2) & ...
```

```
        densitySmooth(2:end-1)>=densitySmooth(3:end))+1;

    if isempty(candidates)
        peakRadii = zeros(0,1);
        return;
    end

    candidates = candidates( ...
        densitySmooth(candidates)>=0.12*max(densitySmooth));

    if isempty(candidates)
        peakRadii = zeros(0,1);
        return;
    end

    % Conserva maximos separados al menos seis bins, ordenados por altura.
    [~,heightOrder] = sort(densitySmooth(candidates),'descend');
    selected = zeros(0,1);

    for j = 1:numel(heightOrder)
        candidate = candidates(heightOrder(j));

        if isempty(selected) || all(abs(candidate-selected)>=6)
            selected(end+1,1) = candidate; %#ok<AGROW>
        end

        if numel(selected)>=4
            break;
        end
    end

    peakRadii = sort(centers(selected));
end

function s = joinNumericVector(v)

    v = v(:).';

    if isempty(v)
        s = "";
        return;
    end

    pieces = strings(size(v));

    for j = 1:numel(v)
        pieces(j) = sprintf('%.9g',v(j));
    end

    s = strjoin(pieces,', ');
end
```

36. Complete console output of the full-data run

36.1. **Coefficient identities, radial audit, branches, towers, blocks, and omitted frequencies.**

Remark 36.1. The console transcript below is included as numerical support. The analytic claims of the manuscript are established in the theorem–proof sections and do not depend on numerical rounding in this transcript.

Listing 2. Complete console output for the full-data run.

```
1
2  ============================================================
3  ZETA PRIME-POWER STRUCTURE: AUDITORIA ANALITICA
4  ============================================================
5  Archivo de ceros       : zeross.bin
6  Ceros disponibles      : 2001052
7  q                      : 2.7182818284590451
8  omega=log(q)           : 1
9  stride                 : 1
10 Qlist                  : 0 3 10 100 1000 10000
11 Puntos analizados       : 2001052
12 gamma inicial           : 14.134725142
13 gamma final             : 1132490.65871441
14
15 ============================================================
16 TEST 2: IDENTIDADES DE COEFICIENTES
17 ============================================================
18 Potencias primas <=Qmax                 : 1280
19 max |(c^-+c^+)-formula|                 : 2.220446e-16
20 max |(c^--c^+)-formula|                 : 2.220446e-16
21 max |c^-/c^+-formula|                   : 1.776357e-15
22       Q      num_terms    Re_AQ          Im_AQ           abs_AQ_minus_half    Re_BQ          Im_BQ
         abs_BQ_minus_half    abs_BQ_plus_half
23     _____    _________    _____    ________________    _________________    _____    ________________
         _________________    _________________
24
25         0         0         0                       0                  0.5      0                       0
                       0.5                 0.5
26         3         2         0      -0.675311086221108    0.840264876793701      0      -0.863660742917475
        0.997952844004547    0.997952844004547
27        10         7         0       -1.07698676185356      1.1873923046777      0       -1.54062594080643
          1.6197309311999      1.6197309311999
28       100        35         0       -1.47655654519435     1.55891604365222      0       -2.87744845271495
         2.92056665700882     2.92056665700882
29      1000       193         0       -1.89626714054451      1.9610785472053      0       -5.31082516388603
         5.33431006985581     5.33431006985581
30     10000      1280         0       -2.55521376172467     2.60367382137378      0       -10.6932626383477
         10.7049458593999     10.7049458593999
31
32
33 ============================================================
34 AUDITORIA RADIAL DIRECTA DE LA BIMODALIDAD
35 ============================================================
36       case_name        mean_radius          std_radius        num_detected_peaks      detected_peak_radii
37     _____________    _________________    _________________    __________________
         ________________________
38
39     "original"       1.92031044763071    0.726547772615351            2             "1.35731059,
        2.49789956"
40     "after_n3"       1.80792479214104    0.308312030736288            1             "1.68017926"
41     "after_n2_n3"    1.78844590851816     0.22140483373138            1             "1.81300495"
42
43 Coeficiente teorico total n=3:
44   c3^- = 0 -1.02370050729i
45   c3^+ = 0 +0.04810295378 51i
46 Q=0      terms=0      c^-res=0.4999989+0.1381673i  c^+res=0.4999989+0.1381673i  c^both=0.4999989+0.1381673i
         Var(F)=1.2420e+00  RMS(E)=1.1145e+00
47 Q=3      terms=2      c^-res=0.4999999+0.0909043i  c^+res=0.4999989+0.1429495i  c^both=0.4999999+0.0956865i
         Var(F)=1.3393e-01  RMS(E)=3.6597e-01
48 Q=10     terms=7      c^-res=0.5000000+0.0636593i  c^+res=0.4999989+0.1500304i  c^both=0.5000000+0.0755224i
         Var(F)=5.9610e-02  RMS(E)=2.4415e-01
49 Q=100    terms=35     c^-res=0.5000000+0.0191267i  c^+res=0.4999989+0.1736326i  c^both=0.5000000+0.0545920i
         Var(F)=2.2040e-02  RMS(E)=1.4846e-01
50 Q=1000   terms=193    c^-res=0.5000000-0.0206343i  c^+res=0.4999990+0.2014465i  c^both=0.5000001+0.0426449i
         Var(F)=1.1282e-02  RMS(E)=1.0622e-01
```

```
Q=10000  terms=1280   c^-res=0.4999999-0.0581400i  c^+res=0.4999989+0.2306199i  c^both=0.4999999+0.0343125i
       Var(F)=8.2559e-03  RMS(E)=9.0862e-02

============================================================
TEST 1: RAMAS; TEST 5: ECUACION FUNCIONAL
============================================================
      Q      num_prime_power_terms    c_after_minus_real    c_after_minus_imag     c_after_plus_real
     c_after_plus_imag    c_after_both_real    c_after_both_imag     distance_c_after_both_to_half
     rms_centered_Y_minus_Pminus    rms_centered_Y_minus_Pplus    rms_centered_Y_minus_both
     rms_after_both_fitted_carrier     energy_Pminus         energy_Pplus
     cross_term_2Re_inner_Pminus_Pplus        F_variance         F_max_abs_deviation_from_mean
     F_block_mean_std        F_anisotropy       carrier_discrete_defect_rms    derivative_defect_rms
    -----    ------------------    ------------------    ------------------    ------------------
     ------------------    ------------------    ------------------    ------------------------
     ---------------------------    -------------------------    -------------------------
     ------------------------------    -----------------    --------------------
     ----------------------------------    --------------------    ------------------------------
     --------------------    -----------------    ---------------------------    ---------------------

       0               0             0.499998912312768       0.138167317526428    0.499998912312768
     0.138167317526428    0.499998912312768     0.138167317526428          0.138167317530709
      1.22927492233399              1.22927492233399              1.22927492233399
     1.11446297144985                        0                     0                                  0
           1.24202771473284           2.53363749717284            0.0101613448739479    0.164704934429685
            0.828969921883291           2.13681541452671
       3               2             0.49999988687813      0.0909043288314701    0.499998893284938
     0.142949512755117    0.499999867850236    0.0956865240601598          0.095686524060251
     0.630307565358949              1.22653992517659             0.626967498686433
     0.365967847866614          1.11258876067543    0.00443650350346841            0.00251726408927867
           0.133932465672121            1.3998447731532           0.00612569360136624    0.670875082815371
             0.476587597924875            1.7823665130755
       10              7             0.49999998247139      0.0636593469198101    0.499998916032037
     0.150030381454066    0.499999986190665     0.075522410847451          0.0755224108474523
     0.568023295911874              1.22456854935429             0.561527861557534
     0.244151425501691          1.18492844347144    0.00871311004796696            0.00421177723983395
           0.0596099185745075           1.05195331714233            0.0042120343511029    0.886135136342853
              0.398840669251055            1.69391674505877
      100              35             0.500000017346496      0.0191267412460307     0.49999893653376
     0.173632564206007     0.50000004156745     0.054591987925598          0.0545919879256138
     0.536555599882361              1.22197814824027             0.524423432995709
     0.148457436196022          1.21916077552125     0.0180717973642778            0.00501426487870665
           0.0220396103618959           0.799206246858361            0.0022297937207615    0.958121438757457
                0.2682174006908           1.45913395069404
     1000             193             0.500000036693381     -0.0206342853236017    0.499998951118377
     0.201446524072374    0.500000075498899    0.0426449212223482          0.042644921222415
     0.526364711348993              1.22166555561873             0.51293359872215
     0.10621775695736          1.23384826802886      0.0252917244776332            0.00469116700005195
          0.0112822118930529          0.668993854159613            0.0011112380638497    0.970189423496525
             0.179876374959657           1.21197809635573
     10000            1280            0.499999922217885     -0.0581399985655881    0.499998861049832
     0.230619861962331    0.499999870954991    0.0343125458703112          0.0343125458705538
     0.523988825848724              1.22334869951959             0.509345817216466
     0.0908622020249825          1.24333446127337      0.030996201483447          -0.000596344803520444
           0.00825593975682874           0.61202824502835           0.000381258918696766
     0.981211983063741         0.150986530997687            1.0471736199134

max error R_j-R_{j-1}=-P_j                 : 8.962228e-16
max error ley de energia de ramas          : 2.220446e-16
max error identidad de proyeccion portadora : 3.559044e-14
max error identidad de constantes           : 5.446388e-14

============================================================
TEST 3: TELESCOPADO EN p,p^2,...,p^R
============================================================
```

```
base_prime    num_powers    largest_power    combined_energy_ratio_sum_over_individual    combined_rms_over_sum_term_rms    boundary_fit_relative_rms_combined    boundary_fit_relative_rms_minus    boundary_fit_relative_rms_plus    boundary_alpha_first_real    boundary_alpha_first_imag    boundary_alpha_last_real    boundary_alpha_last_imag    tail_primitive_start_abs_minus    tail_primitive_start_abs_plus
----------    ----------    -------------    ----------------------------------------    ------------------------------    -----------------------------------    -------------------------------    ------------------------------    -------------------------    -------------------------    ------------------------    ------------------------    ------------------------------    -----------------------------

   89            2             7921                        0.99578526600946    0.951568779802395                  3.25731221427626e-15                1.66529459300868e-14    5.68490958536155e-15             0.999999999999997           5.38298239788951e-18    0.999999999999988        -6.13165759929884e-17             0.022712415828171    0.0146012082752434
   37            2             1369                        0.992325345063945    0.929272152959258                  3.32342530365488e-15                6.90975846079265e-15    2.57414422667853e-14             0.999999999999997          -8.74885970690982e-18    1.00000000000001         -8.01331294276757e-17             0.0386826763307904    0.0223795089575229
   41            2             1681                        0.992798804843908    0.932179807015267                  4.01764974378634e-15                5.71123669041389e-15    6.00240878492108e-15                             1          -4.41766278752356e-18    0.999999999999997         6.86611361451242e-17             0.0362585450928221    0.0212931769572583
   23            2              529                        0.98981553486963    0.914722655159467                  5.29829056019486e-15                3.01909112408219e-14    5.06477325930078e-14             0.999999999999995           1.5256443527521e-17    1.00000000000002         -4.12721429521546e-17             0.052842635596283    0.0281454412919937
   67            2             4489                        0.994867301216485    0.945073421083208                  6.62713306705871e-15                2.92396921729543e-14    3.16762417963749e-14             0.999999999999993           4.06385013591999e-18    0.999999999999977         -6.6991869985725e-17             0.0268592065905553    0.0167695322528614
   73            2             5329                        0.995124247078512    0.947082293993756                  8.02253098923419e-15                4.21876931539434e-14    2.84327775551962e-14             0.999999999999992           2.29352846836488e-17    0.999999999999967         1.25402372809613e-16             0.0255226923919741    0.0160830067737586
   29            2              841                        0.991065442073751    0.921970578232658                  1.69958093184349e-14                3.92022883362443e-15    4.85298540260265e-14               1.00000000000002          5.31348409790742e-18    0.999999999999986        -1.67495027840586e-16             0.0452613167826162    0.0251764129412383
   59            2             3481                        0.994356231992412    0.941904651888199                  1.82233584563793e-14                4.44612460135524e-14     3.3145884657823e-14               1.00000000000002          3.31967944946604e-17    0.999999999999965         2.13895102136555e-16             0.0289902084009457    0.0178408950865232
   61            2             3721                        0.994484921893845    0.942760057482987                  1.89061413493395e-14                4.06345157893362e-14    1.56552838293448e-14             0.999999999999981          -1.56580841679002e-17    0.999999999999966        -2.22783722808597e-17             0.0284133680209016    0.0175536598492679
   43            2             1849                        0.993018430417279    0.933512335891677                  1.91767251608662e-14                9.19221287192386e-15    2.67694821919958e-14               1.00000000000002          -5.47518533793213e-18    0.999999999999988         1.24241990600724e-16             0.0351952587338588    0.0208067465987886
```

```
87         97            2             9409                      0.996034159322401
       0.953412551798485                  2.19836316719303e-14               2.48682193279047e-15
           4.67252549818149e-15             0.999999999999978         -1.96285882336457e-17
       0.999999999999975        -2.20623599182162e-16            0.0216013618259659
       0.0140005339242615
88         71            2             5041                      0.995055469368263
       0.94645716703403                  2.45606582770506e-14               1.40537772877289e-14
          1.35601376253121e-14             1.00000000000002         -2.17658103478551e-17
       1.00000000000007        -3.81439794611675e-16            0.0259471363764858
       0.0163022731943209
89         83            2             6889                      0.995556221672419
       0.950034257670606                  2.80380400407119e-14               7.38122722146823e-15
           1.69133036510308e-14             1.00000000000003         -1.68047878446504e-17
       1.00000000000006        -3.60217187758929e-16            0.0236603972161033
       0.0151069817398424
90         53            2             2809                      0.993923584391844
       0.939166861275507                  3.64856201837887e-14               5.33595501442798e-14
           6.78833363346336e-15             1.00000000000004         -1.38986297087105e-17
       0.999999999999991        -4.48286028808063e-16             0.030939494184714
       0.0187966802009105
91         47            2             2209                       0.99342415440416
       0.935964632967202                  4.02588932254384e-14               4.69976317545872e-15
           3.70119612870561e-14             1.00000000000004          3.64557105503385e-17
       0.999999999999993        -5.00636376597972e-16            0.0333062074556399
       0.0199271852036153
92         79            2             6241                      0.995407211972277
       0.948922007921855                  4.26585420767673e-14               6.84664346328644e-16
           1.95720970351145e-14             0.999999999999957          4.32727341964049e-17
       0.999999999999994         6.93817715660186e-16            0.0243581235750551
       0.015475353217867
93         31            2              961                      0.99142834108407
       0.924010046881153                  4.48560611982839e-14               1.73461849505888e-14
           4.95966705898557e-15             1.00000000000004          1.34960651075476e-18
       1.00000000000001        -1.47682556498033e-16            0.0433336826232223
       0.0243790159002065
94          3            8             6561                      0.994136424160185
       0.925379114250609                  0.0526293055865214                 0.0488679460726364
              0.38539545977793             0.997125200778133          1.04928001769985e-08
       -0.373319182377488          0.00313372076449291              1.09725325128044
       0.080998072339286
95         19            3             6859                      0.988180958947934
       0.896291448276131                   0.099571908742451                 0.0910436270892582
             0.130813323815928             0.994234206127615          8.21480982437346e-07
       0.587653567802991        -6.66788850988072e-05            0.0610579674917809
       0.0314116469586121
96         17            3             4913                      0.987462564808847
       0.891547371679716                   0.104055647033926                 0.0950571818522644
             0.138802345385611             0.993904379532631          7.82474768756128e-08
       0.603159942163849        -1.05608473149603e-05            0.0661990286058359
       0.0331867205015699
97
98 boundary_fit_relative_rms_combined cercano a cero seria evidencia de colapso a los dos bordes del bloque p^
       r.
99
100 ============================================================
101 TEST 4: POSITIVIDAD Y GRAM
102 ============================================================
103       Q            norm_E2          carrier_schur_complement    carrier_projection_abs
       Re_inner_E_Pminus        Re_inner_E_Pplus       cauchy_det_E_Pminus     cauchy_det_E_Pplus
       cauchy_det_Pminus_Pplus    polarization_error_minus    polarization_error_plus
104     -----    --------------------    ------------------------    ----------------------
       --------------------    ----------------------    --------------------    --------------------
       ------------------------    ------------------------    ------------------------
105
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106          0       1.24202771473284          1.24202771473284          4.1753563397406e-15
           0                          0                         0                          0                          0
                                    0                         0
107          3      0.133932465672121         0.133932465672121         3.19449329162395e-14
       -0.00516035229366167      0.00033902164295703 2      0.148985126760466      0.000594076917452501
       0.00493441978003228         2.22044604925031e-16        8.67361737988404e-19
108         10      0.0596099185745075        0.0596099185745074         7.25002166702646e-15
       -0.00596341179707172      0.00020783672099723 2     0.0705979257513303      0.000519344584387362
       0.0103199771600532          2.22044604925031e-16        3.46944695195361e-18
109        100      0.0220396103618959        0.0220396103618958         1.12998339769851e-14
       -0.00697316806144494     -0.00066378017889313 7      0.026821203386263      0.000397854767628292
       0.022026140776441                             0                           0
110       1000      0.0112822118930529        0.0112822118930528         3.76207946660896e-14
       -0.0090127255333809      -0.00296783860935025 0.0138393083795104      0.000276538526093234
       0.0312006486796008          2.22044604925031e-16        1.38777878078145e-17
111      10000     0.00825593975682874       0.00825593975682881         2.86661244710573e-14
       -0.00982808606142384      -0.00476027747414303 0.010168303134064      0.000233242529391743
       0.0385385565639377          2.22044604925031e-16        3.46944695195361e-18
112
113 Gram combinada: min=2.291421e-03, max=1.119809e+00, negativos<-1e-10=0
114 Gram por ramas: min=4.858227e-04, max=1.112574e+00, negativos<-1e-10=0
115
116 ============================================================
117 TEST 5: F_Q POR BLOQUES DE ALTURA, Q=10000
118 ============================================================
119     block      gamma_start          gamma_end           F_mean_real          F_mean_imag
        F_variance         sup_abs_F_minus_global_c    sup_abs_F_minus_block_c     mean_abs_delta_F
        variation_per_height       anisotropy        carrier_discrete_defect_rms    derivative_defect_rms
120     -----    ----------------    ----------------    -----------------    -------------------
        --------------------    ------------------------    ------------------------    -------------------
        ---------------------    -----------------    --------------------------    ---------------------
121
122       1          14.134725142      63699.04754739    0.499994991361003     0.0350523539924386
       0.0054744556993458        0.542532351404617          0.542510726459913        0.0963670633500536
       0.126163323741996       0.979858632599489          0.121147135176878            0.538564667956422
123       2       63699.700151835     118462.736278749    0.500001451274137      0.035105891285452
       0.00701927614984981        0.530144044699919          0.530116241500027         0.110272510195026
       0.167888441917521       0.980474241382446          0.137953328199234            0.726396110177662
124       3      118463.232066064     170651.787459452    0.499998562254379     0.0349210363561646
       0.00745311407766993        0.549893284088027          0.549871555991103         0.113755262822731
       0.181736637024686       0.980873421275848          0.142306171066992            0.7574604344515
125       4       170652.38272511     221291.519378741    0.500000256946116     0.0348042049768492
       0.00767564743723714        0.581090952553066          0.581063883641327         0.116011219222976
       0.191009406027093       0.980878273487911          0.144913294713216            0.880192519706131
126       5      221292.295838797     270836.328162977    0.500000955646856     0.0347080955882823
       0.007831121437243          0.57962624512367           0.579600207778524         0.117381041928519
       0.197536641502951       0.980905060378457          0.146681991054392            0.923600652953341
127       6      270836.931684803     319539.773103185    0.499999624306421     0.0346475825508576
       0.00808884143363537        0.534809090007345          0.53479827792399          0.118850165982451
       0.203463517740728       0.981360633102241          0.14914505740182             1.16913063854399
128       7      319540.570678831     367564.435949345     0.49999957921803     0.0345578903969988
       0.00818709113446676         0.52294466209729          0.52292439742989          0.119910420246151
       0.208180893856153       0.981406498730984          0.150029861940657            0.880453279774671
129       8      367565.006156693     415022.283240227    0.500000487984706     0.0344519123654481
       0.00814253081563704        0.600282121372596          0.600276208543475         0.119491391046747
       0.209930169452775       0.981010897479751          0.149551484403089            1.00807618824186
130       9       415023.15758963     461995.409615676    0.500002125859323     0.0343892982332912
       0.00818298949111592        0.572654690283211          0.572652885861422           0.1200026729604
       0.213007944730247       0.980937348210029          0.150172616772631            1.08111614297039
131      10       461995.761482843     508544.337297334    0.499999128775315     0.0343361215632085
       0.00835629092630199        0.542768812320236          0.542768199088721           0.12137168582189
       0.217396246824282       0.981335496066421          0.152037594340513            0.934750513408792
132      11       508544.765796632     554719.129189251     0.49999902683957     0.0342841671661181
       0.00838655885140866        0.555269896680713          0.55526995649336          0.121722656912353
       0.219791838956823       0.981261093718569          0.152335939237595            1.12844994151658
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133      12       554719.634077261     600557.928334761     0.499999052950246     0.0342521074641879
       0.00848384933678091        0.606087165112338          0.606091467206371          0.12213899474892
       0.222160553553316       0.981322097240396          0.153080139635091             1.07065793214649
134      13       600558.535755246     646093.102175532     0.500000235514038     0.0342171059277988
       0.00852631056710716        0.524846602477766          0.524849827921532         0.122203119197641
       0.223759839331275       0.981196827744373          0.153515460380432             1.14290197565882
135      14        646094.01946502     691351.826509571     0.499999459797555     0.0341425285063966
       0.00852459159575678         0.61202824502835          0.612034526321802         0.122717190800213
       0.226075215930959        0.98130473863714          0.153711972752659             1.52515702182085
136      15       691352.526827545     736357.304575928     0.50000063261387     0.0340996079052541
       0.00863390467854271        0.603869638677834          0.603880275368038         0.123435919580137
       0.228680535306117         0.9813459888377          0.154802894954523             1.13198994875413
137      16       736358.196419715     781127.536440619     0.500000507064813     0.0340763607134163
       0.00863949315690915        0.551934114882366          0.551943379429299         0.123479710718261
       0.229961941722585       0.981411615889858          0.154800438591316             1.02064882439406
138      17       781128.075976077     825679.648395047     0.499999867045727     0.0340594494960409
       0.00867051311819898        0.586536403755823          0.586546166766736         0.123341062550961
       0.230826520207627       0.98119376514941           0.155214716157765             1.15057126782006
139      18       825680.493113046     870029.136314775     0.499999157278646     0.0340083355422251
       0.00880860286374675        0.564338258958265          0.564345300438399         0.124310351932475
       0.233705005485217       0.981676906683931          0.156532439348918             1.09637171176575
140      19       870029.689029286     914187.909332815     0.499999527336173     0.0339700206865135
       0.00876154865120847         0.56850486610471          0.568522179523045         0.124185981373753
       0.234477981944182       0.981561079878874           0.15588197711697            0.995676199295695
141      20       914188.612117913     958168.018537517     0.50000099473625     0.0339345816929237
       0.00878369871741152        0.599171890474147          0.599196831871485         0.124030304858425
       0.235136204413765        0.98142509043805          0.156206512272264             1.06669516660454
142      21        958168.60666595     1001979.98016229     0.499998363067321     0.0338951222311845
       0.0088529271020219         0.597221450310013           0.597233477557632         0.125000771021225
       0.237887755021112       0.981656183204604          0.157057387478477             1.16995875593502
143      22       1001980.68512795     1045632.21311476     0.500003656384172     0.0338638859862856
       0.00874571864686109        0.562103795049353          0.562114143907445         0.124273605310392
       0.237367088718873        0.98131312028741          0.156023921934697             1.01328485216897
144      23       1045632.45556665     1089133.21595891     0.499998860220289     0.0338981896499074
       0.00898518332990823        0.571629074187526          0.571636102485886           0.1254657058582
       0.240474616933205       0.981670300817732          0.158067171395785             1.12663969590805
145      24       1089133.56438868     1132490.65871441     0.500000398441743     0.0338252499519786
       0.00892480514817264        0.590816935924078          0.590833242133321         0.125308756813391
       0.240969628397672       0.981703647581976          0.157535630788163             1.18442415402658
146
147 Rango Re media por bloque : [0.499994991361, 0.500003656384]
148 Rango Im media por bloque : [0.033825249952, 0.0351058912855]
149 Max sup global por bloque : 6.120282e-01
150 Max sup local por bloque  : 6.120345e-01
151
152 ============================================================
153 TEST 6: UNICIDAD Y FRECUENCIAS OMITIDAS
154 ============================================================
155     Qmax     num_omitted_prime_powers_joint_fit    median_minus_amplitude_prime_power
        median_minus_amplitude_null    minus_prime_power_to_null_ratio    median_plus_amplitude_prime_power
         median_plus_amplitude_null    plus_prime_power_to_null_ratio    joint_omitted_explained_fraction
        fit_theory_correlation_real    fit_theory_correlation_imag    fit_theory_correlation_abs
        relative_theory_coefficient_error    F_energy_before_omitted_fit    F_energy_after_omitted_fit
        F_anisotropy        corr_gap_realF        corr_gap_imagF        corr_gap_absF
156     -----    ----------------------------------    ----------------------------------
        --------------------------    ------------------------------    ------------------------------
         --------------------------    ------------------------------    ------------------------------
        --------------------------    --------------------------    --------------------------
        ----------------------------------    --------------------------    --------------------------
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157
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    10000                    56                          0.0011038260045886
     0.000329157644250013              3.35348737564237                  0.000885575009662317
      0.000330982677195599             2.6755932279168                   0.0144445968386994
      0.916473061428038             0.00108815005327049            0.916473707420905
     0.480753428806987               0.00822729788430414           0.00810845788329348
     0.981063724279348    0.187253690050176    0.0933065219921283    -0.139580596671858


Potencias primas omitidas con mayor amplitud:
      n      base_prime    power_exponent        Lambda_n          fit_F_minus_real
     fit_F_minus_imag        fit_F_minus_abs         fit_F_plus_real         fit_F_plus_imag
     fit_F_plus_abs       theory_minus_real     theory_minus_imag      theory_plus_real
     theory_plus_imag      max_branch_fit_abs
    _____    __________    _____________    ________________    ____________________
     ____________________    ___________________    ____________________    ____________________
     ____________________    _________________    ___________________    ________________
     ___________________    ___________________

    10301      10301            1           9.23999625688369     0.000371892244369111
     -0.00161813748988026    0.00166032309433371      0.00033865779123921     0.00138200597601097
     0.00142289480190808            0            -0.00175843054430363           0
     0.0014149869530774    0.00166032309433371
    10169      10169            1           9.22709915579114    -0.000115283050974615
     -0.00156964135711816    0.00157386917239577    -5.07240436672432e-05     0.00133670966276089
     0.0013376717277136            0            -0.00177010677368571           0
     0.00142394668533194    0.00157386917239577
    10181      10181            1            9.2282785171072     0.000278847600421771
     -0.00152895592997706    0.00155417573654751     0.000286682906630887     0.00132354059079306
     0.00135423291365672           0            -0.00176903580369307           0
     0.00142312504251575    0.00155417573654751
    10039      10039            1           9.21423278669153      0.00041374673843581
     -0.00144753771339835    0.00150550715550501     0.000421812244560221     0.00124369263450663
     0.00131327717515647           0            -0.00178183332924528           0
     0.00143294107929247    0.00150550715550501
    10333      10333            1           9.24309793621435     6.14996145285231e-05
     -0.00149251220327071    0.00149377872507917     5.12963409860825e-05     0.00123167205653074
     0.00123273978172087           0            -0.00175563411722169           0
     0.00141284053501554    0.00149377872507917
    10331      10331            1           9.24290436284945    -0.000230890640791369
     -0.00147441795772886     0.0014923869478384    -0.000347632129299537     0.00129628666277216
     0.00134209061147238           0            -0.00175580850836064           0
     0.00141297439683093     0.0014923869478384
    10211      10211            1           9.23122084955554    -0.000633103531925687
     -0.00134320284045877    0.00148492893862073    -0.000631071689296745     0.00106646662668294
     0.00123919431198676           0            -0.00176636674804379           0
     0.00142107721241217    0.00148492893862073
    10369      10369            1           9.24657586455828     4.67547186174683e-05
     -0.00147559974971223    0.00147634028091894     7.99910522234896e-05     0.00124576018232505
     0.00124832567878033           0            -0.00175250381884864           0
     0.00141043758286627    0.00147634028091894
    10453      10453            1           9.25464429753271    -0.000304478868200946
     -0.00141117850295559    0.00144365236410464    -0.000223525782210025        0.00124767153432
     0.00126753620574129           0            -0.00174526357211357           0
     0.0014048785677242    0.00144365236410464
    10271      10271            1           9.23707966916581     0.000151926677381929
     -0.00139593924366338    0.00140418235542959     0.000199437478543235     0.00117395228955498
     0.00119077255846742           0            -0.00176106420552525           0
     0.00141700823205664    0.00140418235542959
    10141      10141            1            9.224341891612    -2.69378011998342e-05
     -0.00139213634163686    0.00139239694011425     1.72266295677679e-05     0.00116438745087312
     0.00116451487432195           0             -0.001772613186228           0
     0.00142586946325406    0.00139239694011425
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    10177       10177             1           9.22788555119193      0.000562556014238549
     -0.00123881678217888    0.00136056469488371      0.000525626337530265      0.00105890848795867
     0.00118218874659523            0            -0.00176939258053915            0
     0.00142339876359955    0.00136056469488371
    10061       10061             1           9.21642184229205     -0.000271184796372421
     -0.0012905239456047    0.00131870893223739     -0.000199026354140983      0.00104720828811089
     0.00106595341752386            0            -0.00177983266568738            0
     0.00143140682870411    0.00131870893223739
    10247       10247             1           9.23474025879972      0.000296226420166288
     -0.00127916398853374    0.00131301584208496      0.000225798114775531      0.00100579696158174
     0.00103083088650042            0            -0.00176317956235639            0
     0.0014186315781825     0.00131301584208496
    10457       10457             1           9.25502688959859      -0.00038302950951308
     -0.00124755231657541    0.00130502811760916     -0.000336465701900575       0.0010635359416545
     0.00111548996756868            0            -0.00174492100357133            0
     0.00140461550806041    0.00130502811760916
    10223       10223             1           9.23239536275701      4.37551338344118e-05
     -0.00126550404521048    0.00126626024188591      6.46513618466203e-05      0.00112056448509538
     0.00112242797713069            0            -0.00176530245864271            0
     0.00142026058016567    0.00126626024188591
    10099       10099             1           9.22019168802656      3.77615682350026e-05
     -0.0012349486588066     0.00123552584996178      0.000122474628046823      0.00095475772183488
     0.000962581083295603           0             -0.00177639258075131            0
     0.00142876845882169    0.00123552584996178
    10429       10429             1           9.25234566612121      0.000104275707715879
     -0.00122341220360067    0.00122784805376671      0.000135594574873729      0.000939541687326359
     0.000949275761282911           0             -0.00174732317276288            0
     0.00140646007086086    0.00122784805376671
    10343       10343             1            9.2440652413778      8.65573758937093e-05
     -0.00120755464889939    0.00121065288559526      0.000151903149928664      0.000998017665518561
     0.00100951167781525            0            -0.00175476293000098            0
     0.00141217180262058    0.00121065288559526
    10111       10111             1            9.2213792190914       9.1842043748573e-05
     -0.00120508425512231    0.00120857892706419      5.50085507262679e-05      0.000972297490544918
     0.00097385232493071            0            -0.00177531032055336            0
     0.00142793835000022    0.00120857892706419


Frecuencias no nulas dominantes de F_Q-mean(F_Q):
    rank    demodulated_frequency       fit_amplitude
    ----    ---------------------    --------------------

       1          7.22212181819358      0.00102998542892402
       2        -0.509252179488009      0.000972670281121882
       3         -7.22212181819358      0.000962787079088038
       4         0.509252179488009      0.000944393206974707
       5         -7.26841747087431      0.000923900818886456
       6         0.694434790210921      0.000912309613268334
       7        -0.694434790210921      0.000907278710300437
       8          7.26841747087431      0.000828631152433364
       9         -2.82403481352441      0.000809343213775063
      10         -6.89805224942848      0.000799201794454691
      11          6.89805224942848      0.000798947722046012
      12          8.42580878789251      0.000771802707211477
      13         0.879617400933833      0.000761874424348131
      14        0.0462956526807281      0.000760539159031102
      15        -0.879617400933833      0.000742831699098273
      16         -5.32400005828373      0.000727744418239324
      17         -8.42580878789251      0.000703624825291971
      18         -8.56469574593469      0.000703431607127999
      19         -8.61099139861542      0.000697831699989827
      20          8.61099139861542      0.000697444147794987
      21          8.93506096738052      0.000693242426624136
      22         -9.02765227274197      0.000691823571238767
      23       -0.0462956526807281      0.000690811206825879
```

```
    24         8.56469574593469     0.000687798036575012
    25         9.02765227274197     0.000680600662464099
    26                        0      0.00067560438002473
    27        -8.33321748253105     0.000673228670039579
    28         3.84253917250043     0.000672369683136021
    29        -3.84253917250043       0.0006713463801668
    30        -6.24991311189829     0.000667528070597226


DIAGNOSTICO FINAL PARA Q=10000
mean(F_Q)                          = 0.499999870955 +0.0343125458703i
|mean(F_Q)-1/2|                   = 3.431255e-02
Var(F_Q)                           = 8.255940e-03
sup |F_Q-mean(F_Q)|                = 6.120282e-01
RMS E_Q                            = 9.086220e-02
razon omitidas potencia/null (-) = 3.353487
razon omitidas potencia/null (+) = 2.675593
fraccion explicada ajuste omitido= 0.014445

============================================================
TERMINADO
Tablas disponibles directamente en el workspace:
  radialAuditTable
  summaryTable
  coefficientCutoffDisplay
  groupSummaryTable
  groupPartialTable
  positivityTable
  blockQmax
  omittedTable
  uniquenessTable
  topFrequencyTable
============================================================
>>
```

37. Conclusion

The finite Möbius interpolation is completely oriented by

$$\Phi_N(s) = M(N) + h_N \frac{R_N(s)}{P_N(s)}. \tag{37.1}$$

Its value at zero, its integer limit, its exponential rate, its rounding criterion, and its recovery of a_N are mutually consistent. The absolutely convergent half-plane does not select $M(N)$ among feasible terminal values.

For the zeta phase walk, the following components are exact:

(i) the endpoint conversion and triangular half identity;
(ii) the positive atomic horizontal defect $D_q(\tau)$;
(iii) the two-branch coefficient algebra;
(iv) the convergent signed invariant $\Delta_\omega(Q) \sim \omega/(\pi^2 \log Q)$;
(v) the raw zero-kernel identity for the carrier coefficient;
(vi) the exact global and local defects (12.15) and (13.9);
(vii) the joint constant–carrier projection formula;
(viii) the projected zero-sampled quadratic form (15.12).

The finite computation shows strong prime-power structure, including the disappearance of the radial double mode after subtraction of $n = 3$, a 99.335% finite variance reduction through $Q = 10000$, stable real carrier coefficient near $1/2$, and detectable omitted prime-power frequencies. None of these finite observations proves uniform energy exhaustion.

The main unresolved statements are not ordinary Parseval estimates. They are uniform estimates for the projected zero-sampled kernel, together with a local explicit-formula limit strong enough to identify the positive atomic measure

$$\sum_{\tau} D_q(\tau)\delta_\tau. \tag{37.2}$$

The termwise area mechanism fails away from the near-height regime. Any stable global relation between the centroid and $\Delta_\omega(Q)$ must therefore be collective.

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